

EM 4300: INORGANIC CHEMISTRY



Larry Mink
CSU San Bernardino

CSU San Bernardino
EM 4300: Inorganic Chemistry

Larry Mink

This text is disseminated via the Open Education Resource (OER) LibreTexts Project (<https://LibreTexts.org>) and like the thousands of other texts available within this powerful platform, it is freely available for reading, printing, and "consuming."

The LibreTexts mission is to bring together students, faculty, and scholars in a collaborative effort to provide an accessible, and comprehensive platform that empowers our community to develop, curate, adapt, and adopt openly licensed resources and technologies; through these efforts we can reduce the financial burden born from traditional educational resource costs, ensuring education is more accessible for students and communities worldwide.

Most, but not all, pages in the library have licenses that may allow individuals to make changes, save, and print this book. Carefully consult the applicable license(s) before pursuing such effects. Instructors can adopt existing LibreTexts texts or Remix them to quickly build course-specific resources to meet the needs of their students. Unlike traditional textbooks, LibreTexts' web based origins allow powerful integration of advanced features and new technologies to support learning.



LibreTexts is the adaptable, user-friendly non-profit open education resource platform that educators trust for creating, customizing, and sharing accessible, interactive textbooks, adaptive homework, and ancillary materials. We collaborate with individuals and organizations to champion open education initiatives, support institutional publishing programs, drive curriculum development projects, and more.

The LibreTexts libraries are Powered by [NICE CXone Expert](#) and was supported by the Department of Education Open Textbook Pilot Project, the California Education Learning Lab, the UC Davis Office of the Provost, the UC Davis Library, the California State University Affordable Learning Solutions Program, and Merlot. This material is based upon work supported by the National Science Foundation under Grant No. 1246120, 1525057, and 1413739.

Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation nor the US Department of Education.

Have questions or comments? For information about adoptions or adaptations contact info@LibreTexts.org or visit our main website at <https://LibreTexts.org>.

This text was compiled on 07/11/2026

TABLE OF CONTENTS

Licensing

1: Atomic Structure

- 1.1: Historical Development of Atomic Theory
- 1.2: The quantum-mechanical model of the atom
- 1.3: Periodic Properties of Atoms
- 1.4: Concept Review Questions Chapter 1
- 1.5: Homework Problems Chapter 1

2: Symmetry and Group Theory

- 2.1: Symmetry Elements and Operations
- 2.2: Point Groups
- 2.3: Matrix Representations of Symmetry Operations and Character Tables
- 2.4: Concept Review Questions Chapter 2
- 2.5: Homework Problems Chapter 2

3: Review of Chemical Bonding

- 3.1: Prelude to Chemical Bonding
- 3.2: Valence Bond Theory- Lewis Dot Structures, the Octet Rule, Formal Charge, Resonance, and the Isoelectronic Principle
- 3.3: The Shapes of Molecules (VSEPR Theory) and Orbital Hybridization
- 3.4: Bond Polarity and Bond Strength
- 3.5: Discussion Questions
- 3.6: Problems
- 3.7: References

4: Molecular Orbital Theory

- 4.1: Prelude to Molecular Orbital Theory
- 4.2: Constructing Molecular Orbitals from Atomic Orbitals
- 4.3: Orbital Symmetry
- 4.4: σ , π , and δ orbitals
- 4.5: Diatomic Molecules
- 4.6: Orbital Filling
- 4.7: Periodic Trends in π Bonding
- 4.8: Three-center Bonding
- 4.9: Building up the MOs of More Complex Molecules- NH_3 , P_4
- 4.10: Homology of σ and π orbitals in MO diagrams
- 4.11: Chains and Rings of π -Conjugated Systems
- 4.12: Discussion Questions
- 4.13: Problems
- 4.14: References

5: Coordination Chemistry I - Structures and Isomers

- 5.1: History
- 5.2: Nomenclature
- 5.3: Coordination Numbers and Structures

- 5.4: Isomerism
- 5.5: Concept Review Questions Chapter 5
- 5.6: Homework Problems Chapter 5

6: Coordination Chemistry III - Bonding

- 6.1: 7.1 Bonding in Coordination Compounds
- 6.2: Concept Review Questions Chapter 7
- 6.3: Homework Problems Chapter 7

7: Metals and Alloys- Structure, Bonding, Electronic and Magnetic Properties

- 7.1: Prelude to Metals and Alloys
- 7.2: Unit Cells and Crystal Structures
- 7.3: Bravais Lattices
- 7.4: Crystal Structures of Metals
- 7.5: Bonding in Metals
- 7.6: Conduction in Metals
- 7.7: Atomic Orbitals and Magnetism
- 7.8: Ferro-, Ferri- and Antiferromagnetism
- 7.9: Hard and Soft Magnets
- 7.10: Discussion Questions
- 7.11: Problems
- 7.12: References

8: Metals and Alloys - Mechanical Properties

- 8.1: Defects in Metallic Crystals
- 8.2: Work Hardening, Alloying, and Annealing
- 8.3: Malleability of Metals and Alloys
- 8.4: Iron and Steel
- 8.5: Amorphous Alloys
- 8.6: Discussion Questions
- 8.7: Problems
- 8.8: References

9: Ionic and Covalent Solids - Structures

- 9.1: Prelude to Ionic and Covalent Solids - Structures
- 9.2: Close-packing and Interstitial Sites
- 9.3: Structures Related to NaCl and NiAs
- 9.4: Tetrahedral Structures
- 9.5: Layered Structures and Intercalation Reactions
- 9.6: Bonding in TiS_2 , MoS_2 , and Pyrite Structures
- 9.7: Spinel, Perovskite, and Rutile Structures
- 9.8: Discussion Questions
- 9.9: Problems
- 9.10: References

10: Ionic and Covalent Solids - Energetics

- 10.1: Ionic Radii and Radius Ratios
- 10.2: Structure Maps

- 10.3: Energetics of Crystalline Solids- The Ionic Model
- 10.4: Born-Haber Cycles for NaCl and Silver Halides
- 10.5: Kapustinskii Equation
- 10.6: Discovery of Noble Gas Compounds
- 10.7: Stabilization of High and Low Oxidation States
- 10.8: Alkalides and Electrides
- 10.9: Resonance Energy of Metals
- 10.10: Prelude to Ionic and Covalent Solids - Energetics
- 10.11: The Strange Case of the Alkali Oxides
- 10.12: Lattice Energies and Solubility
- 10.13: Discussion Questions
- 10.14: Problems
- 10.15: References

11: Electronic Properties of Materials - Superconductors and Semiconductors

- 11.1: Prelude to Electronic Properties of Materials - Superconductors and Semiconductors
- 11.2: Metal-Insulator Transitions
- 11.3: Superconductors
- 11.4: Periodic Trends- Metals, Semiconductors, and Insulators
- 11.5: Semiconductors- Band Gaps, Colors, Conductivity and Doping
- 11.6: Semiconductor p-n Junctions
- 11.7: Diodes, LEDs and Solar Cells
- 11.8: Amorphous Semiconductors
- 11.9: Discussion Questions
- 11.10: Problems
- 11.11: References

12: Physical Techniques in Inorganic Chemistry

13: Coordination Chemistry IV - Electronic Spectra

- 13.1: Microstates and Terms
- 13.2: Term splitting in ligand fields, selection rules, Tanabe-Sugano diagrams. Metal to ligand, and ligand to metal transitions
- 13.3: Concept Review Questions Chapter 8
- 13.4: Homework Problems Chapter 8

14: Coordination Chemistry V - Reactions and Mechanisms

- 14.1: Substitution Reactions and their Mechanisms
- 14.2: Concept Review Questions Chapter 9
- 14.3: Homework Problems Chapter 9

15: Acid-Base Chemistry

- 15.1: Prelude to Acid-Base Chemistry
- 15.2: Brønsted and Lewis Acids and Bases
- 15.3: Hard and Soft Acids and Bases
- 15.4: The Electrostatic-Covalent (ECW) Model for Acid-Base Reactions
- 15.5: Frustrated Lewis Pairs
- 15.6: Discussion Questions

- [15.7: Problems](#)
- [15.8: References](#)

[16: Redox Stability and Redox Reactions](#)

- [16.1: Prelude to Redox Stability and Redox Reactions](#)
- [16.2: Balancing Redox Reactions](#)
- [16.3: Electrochemical Potentials](#)
- [16.4: Latimer and Frost Diagrams](#)
- [16.5: Redox Reactions with Coupled Equilibria](#)
- [16.6: Pourbaix Diagrams](#)
- [16.7: Discussion Questions](#)
- [16.8: Problems](#)
- [16.9: References](#)

[17: Transition-Metal Storage, Transport, and Biomineralization](#)

- [17.1: Biological Significance of Iron, Zinc, Copper, Molybdenum, Cobalt, Chromium, Vanadium, and Nickel](#)
- [17.2: Biological Systems of Metal Storage](#)
- [17.3: Chemical Properties Relative to Storage and Transport](#)
- [17.4: Iron Biomineralization](#)
- [17.5: Transport of Iron](#)
- [17.6: Transport of Zinc, Copper, Vanadium, Chromium, Molybdenum, and Cobalt](#)

[Index](#)

[Glossary](#)

[Detailed Licensing](#)

Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

1: Atomic Structure

- [1.1: Historical Development of Atomic Theory](#)
- [1.2: The quantum-mechanical model of the atom](#)
- [1.3: Periodic Properties of Atoms](#)
- [1.4: Concept Review Questions Chapter 1](#)
- [1.5: Homework Problems Chapter 1](#)

Dr. Kai Landskron ([Lehigh University](#)). If you like this textbook, please consider to make a donation to support the author's research at Lehigh University: [Click Here to Donate](#).

This page titled [1: Atomic Structure](#) is shared under a [not declared](#) license and was authored, remixed, and/or curated by [Kai Landskron](#).

1.1: Historical Development of Atomic Theory

Development of Atomic Theory

Since the ancient times, humans thought of explaining the material world in all of its complexity. The basic idea behind all element theories is that the matter that surrounds us is composed of more simple matter. This matter may be composed of even simpler matter. The most simple matter would be called an element.

The first element theories were one element theories: The greek philosopher Thales thought water was the only element, and everything was a form of water. Anaximenes thought air was the only element, and Heraclitus believed fire was the only element (Figure 1.1.1).



Figure 1.1.1 Thales (640-545 BC) (Attribution: Ernst Wallis et al [Public domain] https://commons.wikimedia.org/wiki/File:Illustrerad_Verldshistoria_band_I_Ill_107.jpg), Anaximenes (585-525 BC) (Attribution: https://commons.wikimedia.org/wiki/File:Anaximenes_of_Miletus_Painting.jpg) and Heraclitus (544 – 483 BC) (Attribution: RoyFokker [CC BY-SA (<https://creativecommons.org/licenses/by-sa/4.0>)]), respectively.

However, with only one element assumed it was difficult to explain the material world in all its complexity satisfactorily. This could be done more satisfactorily by a multi-element theory. The first philosopher who introduced a multi-element theory was Empedocles. He suggested a four-element theory with fire, air, water, and soil as the elements (Figure 1.1.2).



Figure 1.1.2 Empedocles (495 – 435 BC) (Attribution: Thomas Stanley, The history of philosophy, 1655 [Public domain] <https://commons.wikimedia.org/wiki/File:Philosophy.jpg>)

First atomistic ideas



Figure 1.1.3 Democritus (460 – 370 BC) (Attribution: Strannik 92 [CC BY-SA (<https://creativecommons.org/licenses/by-sa/3.0/>)] <https://commons.wikimedia.org/wiki/File:Demokrit.jpg>)

The element theories of Thales, Anaximenes, Heraklitus, and Empedocles were all non-atomistic. This means that they did not include the idea that elements were made of small particles that were indivisible. The first greek philosophers that introduced atomistic element theory were Leukippes, and Democritus (Figure 1.1.3). They assumed that particles cannot be divided into smaller particles infinitely often. Ultimately, after many divisions, one would arrive at particles that could not be further divided, and these particles would be called atoms. Atomistic element theory allows for many different elements which helps to explain the complexity of the material world satisfactorily. However, Leukippes and Democritus did not know how many different elements there were, and how different atoms of different elements were actually distinguished. This question would not be answered until about 2000 years later.

Modern Atomic Theories



Figure 1.1.4 Antoine Lavoisier, the father of modern chemistry, discovered the law of conservation of mass in 1774. (Attribution: Louis Jean Désiré Delaistre (1800–1871); drawing by Julien Léopold Boilly (1796–1874). [Public domain], https://commons.wikimedia.org/wiki/File:Lavoisier_color.jpg)

Although the atomistic idea was already known in the antique, it became forgotten for a long time and was only reintroduced about 2000 years later with Dalton's atom hypothesis. What led to Dalton's atom hypothesis? The first discovery that was important to the development of modern atomic theory was the law of the conservation of mass by Antoine Lavoisier (Fig. 1.1.4). Historically, the conservation of mass and weight was kept obscure for millennia by the buoyant effect of the Earth's atmosphere on the weight of gases, an effect not understood until the vacuum pump first allowed the effective weighing of gases using scales. Until then, in many instances mass seemed to appear or disappear. For example, the mass of wood seems to disappear when it is burned. However, the mass of the wood actually does not disappear it is just converted into the mass of gases, mainly carbon dioxide. When scientists realized that mass never disappeared they could for the first time embark on quantitative studies of the transformations of substances. This in turn led to the idea of chemical elements, as well as the idea that all chemical processes and transformations are simple reactions between these elements. The **law of conservation of mass** states that the mass of a closed system of substances will remain constant, regardless of the processes acting inside the system. An equivalent statement is that matter cannot be created or destroyed, although it may change form. This implies that for any chemical process in a closed system, the mass of the reactants must equal the mass of the products.

Law of Constant Composition

The discovery of the law of the conservation of mass led to the discovery of the law of the constant composition (also called law of definitive proportions) by Joseph Proust (Fig. 1.1.5). This law was the result of chemical analysis that determined the mass ratio of elements in pure substances. It was found that a pure substance always contains exactly the same proportion of elements by mass. For example the element analysis of pure substances containing the elements carbon and oxygen would be either 42.9% carbon and 57.1% oxygen or 27.3% carbon and 72.7% oxygen.



Figure 1.1.5 Joseph Proust (1754-1826) (Attribution: The original uploader was HappyApple at English Wikipedia. [Public domain] https://commons.wikimedia.org/wiki/File:ust_joseph.jpg)

The law was questioned by Proust's fellow Frenchman Claude Louis Berthollet, who argued that the elements could combine in any proportion. The very existence of this debate was because at the time the distinction between pure chemical compounds and mixtures had not yet been fully developed. When two pure compounds of two different elements are mixed, then the mixture can have a continuous mass ratio between the two elements because compounds can be mixed at any ratio. So, for the example in the two compounds between carbon and oxygen, in mixtures of the two there could be carbon to oxygen ratios varying continuously between 27.7 % to 42.9% carbon, and 42.9% to 72.7% oxygen. It was an accomplishment of Proust that he was able to correctly distinguish between pure compounds and mixtures.

The Law of Multiple Proportions



Figure 1.1.6 John Dalton 1766-1844 (Attribution: Charles Turner [Public domain] https://commons.wikimedia.org/wiki/File:les_Turner.jpg)

John Dalton (Fig. 1.1.6) discovered the law of multiple proportions by close analysis of Proust's law of constant composition. He noticed that if two elements form more than one compound between them, then the ratios of the masses of the second element which combines with a fixed mass of the first element will be the ratios of small whole numbers, also called integer numbers. Let us see what is the meaning of the law of the multiple proportions by an example:

Example 1.1.1

The first compound contains 42.9% by mass carbon and 57.1% by mass oxygen.

The second compound contains 27.3% by mass carbon and 72.7% by mass oxygen.

Show that the data are consistent with the Law of Multiple Proportions!

Solution

In 100 g of the first compound (100 is chosen to make calculations easier) there are 57.1 g O and 42.9 g C.

The mass of O per gram C is:

$$57.1 \text{ g O} / 42.9 \text{ g C} = 1.33 \text{ g O per g C}$$

In 100 g of the second compound, there are 72.7 g O and 27.3 g C.

The mass of oxygen per gram of carbon is:

$$72.7 \text{ g O} / 27.3 \text{ g C} = 2.66 \text{ g O per g C}$$

Dividing the mass O per g C of the second (larger value) compound by the mass O per g C of the first compound gives:

$$2.66 : 1.33 = 2 : 1$$

2 and 1 are both integer numbers.

Dalton's Atom Hypothesis

Dalton argued that atoms could explain the law of the multiple proportions. If one assumed that elements were made of the same type of indivisible particles, that are identical in mass and all properties then, because these particles can only come in integer numbers, different atoms can be combined to form compounds also only in integer numbers. Thus, in a pure compound atoms of unlike elements would be combined in small whole number ratios. Consequently a given compound always has the same relative number and types of atoms. Chemical reactions would involve reorganization of atoms, but the atoms would retain their identity. Atoms could be rearranged in a chemical reaction but not created or destroyed.

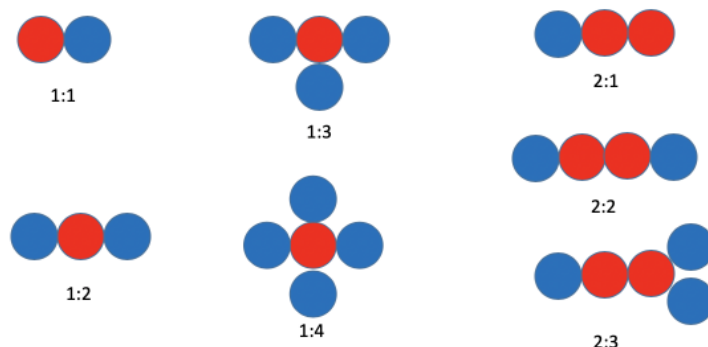


Figure 1.1.7 Illustration of the law of the multiple proportions for two generic, different atoms (red and blue balls).

Figure 1.1.7 is a simple illustration how atoms can explain the law of multiple proportions. The blue and red balls symbolize different atoms of different elements. It is possible to combine a red atom with one, two, three, or four blue atoms to make four different compounds. It is for example also possible to combine two red atoms with one, two or three blue atoms. In all of these compounds as well as in all other thinkable ones, the blue and red atoms can only be combined in integer numbers.

Dalton's Atom Symbols

Dalton also thought of symbols for the atoms of the different elements. The symbols are different from the element symbols that are used today, but the concept is the same. You can see some of them in Figure 1.1.8. For example oxygen is represented by a white ball, while carbon is symbolized by a black ball. You can also see that Dalton already combined atom symbols of different elements to illustrate compounds. For example he combined one white ball with one black ball to indicate carbon monoxide, while he combined two white balls with one black ball to indicate carbon dioxide. In this case he correctly identified the ratios of atoms in the two compounds. However, you can see that this is not always the case. For example, he combined one hydrogen atom (white ball with a dot in the middle) with one oxygen atom to indicate the composition of water. However, as we know today, a water molecule has two hydrogen atoms and one oxygen atom. Why has he been right with the carbon-oxygen compounds, but wrong about the water? The answer is that at the time he could determine the atom ratios only from the law of the multiple proportions. At the time there was only one compound known that was made of hydrogen and oxygen, water, so he assumed the simplest atom ratio of 1:1. If he had known hydrogen peroxide, which has the composition H_2O_2 , and thus half as much hydrogen per oxygen, he would have probably correctly assigned the correct hydrogen to oxygen ratio to water. In the case of the carbon oxides, he assigned the compositions correctly because both carbon monoxide and carbon dioxide were known at the time.

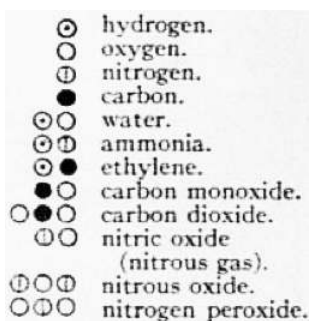


Figure 1.1.8 Dalton's atom symbols (Attribution: Encyclopædia Britannica, 1911 [Public domain]
https://commons.wikimedia.org/wiki/File:an_symbols.png)

Subatomic Particles

Dalton's atomic theory stated that different elements were made of different atoms, but did not explain what made the atoms different. The answer is that atoms are composed of subatomic particles, protons, neutron, and electrons, and the number of the protons and electrons in an atom defines the element it represents. So after all, atoms are not so indivisible. However, an atom can still be seen as the smallest particle that represents the full properties of an element. If an atoms gets divided further into subatomic particles then these properties are being lost. To understand atoms further, we must therefore understand the subatomic particles they are made of. The first subatomic particle that was discovered was the electron. It was found through the investigation of so-called cathode rays which were a mysterious phenomenon at the time. Cathode rays occur when a high voltage of about 10-20 kV is applied to an evacuated tube (0.0001 mm Hg). You can see such a tube with a cathode ray going through in Figure 1.1.9. The big question was: What causes these rays?

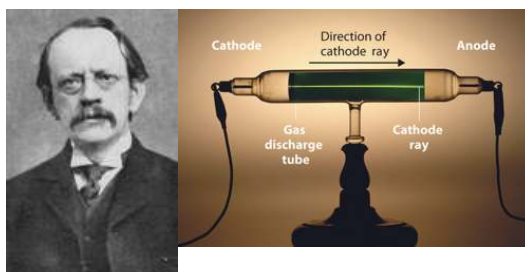


Figure 1.1.9 Joseph John Thomson (1856-1940) (Attribution: https://commons.wikimedia.org/wiki/File:J_Thomson.jpg, PD-US-expired) and his experimental apparatus (Attribution: Chemlibetexts  <https://chem.libretexts.org/@api/dek...338&height=228,creativecommons.org/licenses/by-nc-sa/3.0/us/>).

The answer was found by John Joseph Thomson. He applied electric and magnetic fields to these rays and found them deflected. When an electric field was applied the beam was deflected toward the positive pole. When an equally strong electric and magnetic field was applied, and the magnetic field was perpendicular to the electric field then these fields canceled out, and the rays were no longer deflected. Thomson concluded that these rays must consist of negatively charged particles emitted by the negatively charged electrode because they were attracted by the positive pole of the electric field: Electrons. Knowing the forces associated with the electrical and magnetic fields, he was able to calculate a mass to charge ratio of the electrons which is $5.6857 \times 10^{-9} \text{ g/C}$. After Thomson, Robert Millikan performed another experiment, the so-called oil drop experiment, that allowed to determine the charge and the mass of the electron. The charge of an electron is $-1.6 \times 10^{-19} \text{ C}$, and its mass is $9.1 \times 10^{-28} \text{ g}$.

Thomson's plum pudding atomic model

The discovery of the electron led Thomson to the development of a first atomic model that would include a subatomic particle. It is named plum pudding model (Fig. 1.1.10). In the Thomson atom model electrons are embedded as little particles in a positively charged mass like raisins are embedded in a cake. This atom model is still far from the atom model that we accept today, but it represented an important step forward, because it introduced for the first time the idea that atoms are not indivisible, but contain subatomic particles.

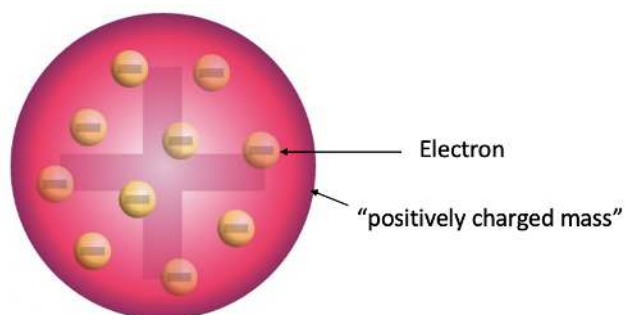


Figure 1.1.10 Thomson's plum pudding atomic model (Attribution: Fatfission (public domain) https://commons.wikimedia.org/wiki/File:Plum_pudding_atom.svg)

Rutherford's Atom Model

Ernest Rutherford (Fig. 1.1.11) wanted to test Thomson's plum pudding atom model by experiment. To do so, he placed a photographic film around a gold foil (Figure 1.1.12). In the following he bombarded the gold foil with alpha radiation. Alpha radiation is a form of radioactivity and consists of helium nuclei, also called alpha-particles. It was known at this time the alpha-particles were positively charged and had a large mass compared to an electron. Rutherford's hypothesis was that if Thomson's atom model was true, then the alpha particles should all go straight through the gold foil like a bullet going through a plum pudding. Therefore, a blackening of the photographic film, which served as the alpha radiation detector, should only occur directly behind the gold foil.



Figure 1.1.11 Sir Ernest Rutherford (1871-1931) who earned the Nobel prize for chemistry 1908. (Attribution: George Grantham Bain Collection (Library of Congress) [Public domain], commons.wikimedia.org/wiki/F...erford_LOC.jpg)

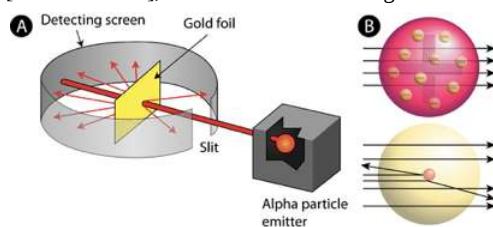


Figure 1.1.12 Rutherford's Gold Foil Experiment

However, what Rutherford observed was that while most of the alpha-particles went straight through the gold foil, some of them were deflected, and a few of them were even reflected. Rutherford concluded that Thomson's plum pudding model must be wrong. A bullet shot at a plum pudding would just never be reflected by the plum pudding and come back at you.

He suggested that the atom would be made of a positively charged nucleus where almost the entire mass of the atom would be concentrated. In the event of a collision between the alpha particle and that nucleus, the alpha particle would be reflected. In the event of an alpha particle passing by the nucleus close, the alpha particle would be deflected. In the event of an alpha particle passing the nucleus in larger distance, the alpha particle would pass the atom practically non-deflected. The observation that most of the alpha particles were not deflected led Rutherford to the conclusion that the overall atom must be much larger than the nucleus, in fact, the data allowed him to calculate that the radius of the atom would be about 10,000 times larger than the radius of the nucleus. This implies that the atoms would be mostly empty space. To explain the empty space, Rutherford assumed that the electrons would move in orbits around the nucleus like planets around the sun. This would be called the planetary model of the atom (Fig. 1.1.13).

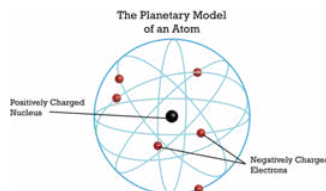


Figure 1.1.13 Rutherford's Atom Model

Bohr's Atom Model

Rutherford's atom model was another big step forward in the development of atomic theory, however there were inherent problems with it as it violated fundamental principles of physics. An electron in an orbit is a self-accelerating electrically charged particle, and according to the laws of physics such particles must emit electromagnetic radiation. However, under normal circumstances atoms do not emit electromagnetic radiation. Secondly, even if the electron emitted electromagnetic radiation, then this would mean that the electron would lose energy because electromagnetic radiation is a form of energy. However, an electron constantly losing energy would make it unstable in its orbit around the nucleus, and it should spiral downward closer to the nucleus until the atom

was eventually collapsed. It is however, experimentally not observed that atoms collapse, they are quite stable species. These difficulties of the Rutherford atom model meant that it could not be the final answer to atomic structure. Niels Bohr (Fig. 1.1.14) was aware of the problems of the Rutherford model, and two new developments in physics, namely the concept of the quantization of energy and atomic spectra helped him to develop an improved atom model, known as Bohr model. To understand this model let us look first at the quantization of energy and atomic spectra.



Figure 1.1.14 Niels Bohr (1885 - 1962) (Attribution: Bain News Service, publisher Restored by: Bammesk [Public domain], commons.wikimedia.org/wiki/F...in_-_35303.jpg)

Blackbody Radiation

The quantization of energy was discovered in the context of the physical phenomenon called “blackbody radiation”. Blackbody radiation is the electromagnetic radiation any object sends out due to its temperature (Fig. 1.1.15).



Figure 1.1.15 An example of blackbody radiation emitted by hot, volcanic lava. (Attribution: Hawaii Volcano Observatory (DAS) [Public domain], commons.wikimedia.org/wiki/F...hoehoe_toe.jpg)

Its distribution of wavelengths follows curves that depend on the temperature and are shown in Figure 1.1.16. You can see that for each temperature the intensity first increases with increasing wavelength, then goes through a maximum, and finally decreases again. You can also see that the overall intensity of the blackbody radiation increases with temperature and that the maximum of the curve shifts to smaller wavelengths with increasing temperature. Objects at room temperature do not emit much blackbody radiation and the wavelengths are far longer than visible for the human eye. However, when an object is heated high enough, intensity increases and wavelength decreases, and the object starts to glow. For instance, lava (Figure 1.1.15) has a temperature high enough so that its blackbody radiation is visible for the human eye. With increasing temperature objects first glow red, then orange, then yellow, and eventually white. This is because red is the color associated with the largest wavelength, and as temperature increases other colors mix into red, until the object glows white. At extremely high temperatures the blackbody radiation also has a significant intensity in the UV region. Such temperatures occur in welding processes, for example, and for this reason welders need to wear glasses that block the UV radiation. The wavelength and intensity of blackbody radiation can be easily measured, but at the beginning of the 20th century there was no good explanation for its behavior. Classical theory predicted that intensity would continuously increase with decreasing wavelength at any temperature which was not in accordance with experimental observation.

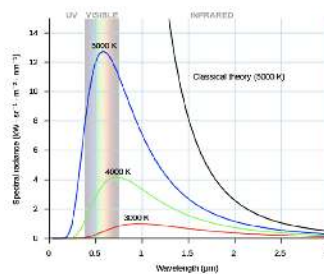


Figure 1.1.16 Experimentally measured intensity of blackbody radiation as a function of wavelength for 3000, 4000, and 5000 K vs. intensity predicted by classical theory (5000 K). (Attribution: Darth Kule commons.wikimedia.org/wiki/F...Black_body.svg)

To bring experiment and theory in accordance, Max Planck made the radical assumption that the energy associated with radiation of a given wavelength or a given frequency was quantized. It would be an integer multiple n of that frequency ν , multiplied with proportionality constant h , now known today as the Planck constant.

$$E = n h \nu$$

Equation 1.1.1 The Planck-Einstein equation.

Using this assumption he was able to derive the Planck equation (Equation 1.1.2) which correctly describes the intensity and wavelength distribution of the blackbody radiation for any temperature. The correct description of blackbody radiation strongly supported Planck's assumption that energy was quantized, but it did not prove it or explain it. The proof and the explanation was found only later. Initially Planck assumed it merely to fit the data.

$$E(\lambda, T) = \frac{2hc^2}{\lambda^5} * \frac{1}{e^{\left(\frac{hc}{\lambda kT}\right)} - 1}$$

h = Planck's constant = $6.626 * 10^{-34} \text{ J} * \text{s}$
 c = speed of light = $2.997925 * 10^8 \text{ m / sec}$
 λ = wavelength (m)
 k = Boltzmann's constant = $1.381 * 10^{-23} \text{ J/K}$
 T = temperature (K)

Equation 1.1.2 Planck's equation

Bohr's Atom Model

The second development that contributed to Bohr's atom model was the absorption and emission spectra of atoms. It was experimentally observed that under certain circumstances atoms would send out or absorb electromagnetic radiation of discrete wavelengths that were characteristic for an atom. For example, H atoms would absorb or emit at four discrete wavelengths in the visible region of the electromagnetic spectrum (Fig. 1.1.17).



Figure 1.1.17 The hydrogen emission spectrum in the visible region (Balmer series). (Attribution: Merikanto, Adrignola [CC0], commons.wikimedia.org/wiki/File:spectrum-H.svg)

This is known as the Balmer series named after its discoverer Johann Balmer. Friedrich Paschen and Chester Lyman later found that H also absorbs and emits at discrete wavelengths in the IR and UV region respectively. These wavelengths are known as the Paschen and Lyman series, respectively (Fig. 1.1.18).



Figure 1.1.18 Friedrich Paschen (1865 – 1947) (Attribution: www.maerkischeallgemeine.de [Public domain], commons.wikimedia.org/wiki/File:n_Physiker.jpg) and Chester Lyman (1814-1988) (Attribution: commons.wikimedia.org/wiki/File:mith_Lyman.jpg), respectively.

For the absorption and emission spectra of H a simple mathematical relationship between the energies associated with the wavelengths can be found. The energy is proportional to a constant, today known as the Rydberg constant, times one over an integer number square minus one over another integer number square, whereby the second integer number would be larger than the first one (Eq. 1.1.3). The observation of integer numbers showed that also the energy and the wavelengths of the atomic absorption and emission spectra are quantized. The question was what the quantization of the absorption spectra meant for atomic structure.

$$\Delta E = R_H \left(\frac{1}{n_{low}^2} - \frac{1}{n_{high}^2} \right)$$

Equation 1.1.3 Rydberg formula for the emission of light with particular energies.

Bohr answered the question the following way. He argued, that like in the Rutherford model the electrons would move in orbits around the nucleus. A balance of opposite centrifugal forces and Coulomb attractions would hold the electron stable in the orbit. However, because energy is quantized, also the angular momentum of the electron would be quantized, and thus, only discrete radii would be allowed. The most inner orbit would have the quantum number $n=1$, the next higher orbit the quantum number $n=2$, and so forth (Fig. 1.1.19).

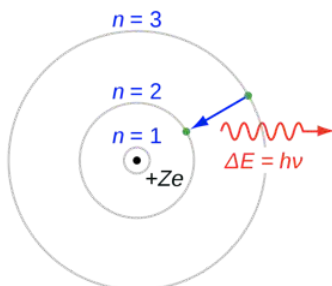


Figure 1.1.19 Bohr atom model (Attribution: JabberWok [CC BY-SA (<http://creativecommons.org/licenses/by-sa/3.0/>)], commons.wikimedia.org/wiki/F...atom_model.svg)

The quantization of the electron energies and the radii would be the explanation why electrons, despite self-accelerating, do not continuously emit electromagnetic radiation. Electromagnetic radiation is only emitted when an electron jumps from an outer orbit of higher energy to an inner orbit of lower energy. This radiation must have a discrete wavelength because the energy difference between two orbits is discrete. Vice versa, an atom can absorb electromagnetic radiation of specific wavelength and energy that is suitable to make the electron jump from an inner to an outer orbit. In sum, the quantization of the energy and the radii would explain the quantization of the absorption and emission spectra. The question is: Can the radii be calculated, and what are the associated energies of the electrons in the orbits?

Definition: Bohr's Postulates (1913)

1. An electron in an atom moves in a circular orbit about the nucleus under the influence of the Coulomb attraction between the electron and the nucleus, obeying the laws of classical mechanics.
2. Instead of the infinity of orbits which would be possible in classical mechanics, it is only possible for an electron to move in an orbit for which its orbital angular momentum L is an integral multiple of $h/2\pi$.
3. Despite the fact that it is constantly accelerating, an electron moving in such an allowed orbit does not radiate electromagnetic energy. Thus, its total energy E remains constant.
4. Electromagnetic radiation is emitted if an electron, initially moving in an orbit of total energy, discontinuously changes its motion so that it moves in an other orbit of total energy. The frequency of the emitted radiation is equal to the quantity divided by h .

To calculate the radii and energies of the electron in the H atom Bohr made two assumptions: Firstly, the centrifugal force associated with the electron moving in the orbit would be assumed equal to the Coulomb force between the electron and the proton so that the electron would be stable in the orbit. Secondly, the angular momentum of the electron would be quantized and an integer multiple of the Planck constant h . The factor 2π is because $E = h/2\pi \times$ angular frequency. The angular frequency (also called angular speed) is the angular displacement (in degree or rad) per time unit. Angular frequency is frequency $\times 2\pi$. We can rearrange this equation by solving it for v and then insert the equation into equation I giving the result as shown in Fig. 1.1.20.

I.

$$F_{cent} = F_{Coulomb}$$

$$\frac{mv^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2}$$

II.

$$2\pi r m v = nh$$

Solve for v:

$$v = \frac{nh}{2\pi r m}$$

↓ insert II. in I.

$$\frac{mn^2 h^2}{4\pi^2 r^3 m^2} = \frac{e^2}{4\pi\epsilon_0 r^2}$$

v = velocity of the electron
 e = elementary charge = $1.602176\ 53(14) \times 10^{-19}$ C
 ϵ_0 = dielectric constant of the vacuum
 m = mass of the electron

Figure 1.1.20 Derivation of the radii of the electron orbits (part 1).

We can then multiply this equation by $4\pi\epsilon_0 r^2$, divide by e^2 , and multiply it by r . This gives us the term $r = (n^2 h^2 \epsilon_0) / (\pi m e^2)$. Analyzing the term shows us that r is only a function of the quantum number n , specifically the radius is proportional to n^2 (Fig. 1.1.21). All other terms are constants, namely the Planck constant h , the dielectric constant of the vacuum, π , the mass of the electron, and the elementary charge. By inserting the quantum numbers into the equation we can calculate the actual values for the radii. For example, when we insert 1 for the quantum number n , we get the radius for the most inner orbit of the electron in the H atom. It is 5.29×10^{-11} m. If we inserted 2 for n , we would get the second radius, if we inserted 3, we would get the third radius and so on.

$$\frac{mn^2 h^2}{4\pi^2 r^3 m^2} = \frac{e^2}{4\pi\epsilon_0 r^2} \quad | \times 4\pi\epsilon_0 r^2$$

$$\frac{n^2 h^2 \epsilon_0}{\pi r m} = e^2 \quad | \div e^2 \quad | \times r$$

$$r = \frac{n^2 h^2 \epsilon_0}{\pi m e^2}$$

For the first radius ($n=1$): $r = \frac{1^2 h^2 \epsilon_0}{\pi m e^2} = 5.29 \times 10^{-11} \text{ m}$

Figure 1.1.21 Continuation of the derivation of the radii of the electron orbits.

Now let us calculate the energies of the electron on these radii. Generally, the overall energy of the electron is the sum of the kinetic and potential energies. The kinetic energy of a moving object is given by $E_{kin} = 1/2 mv^2$. We know that m must be the mass of the electron, but what is the velocity of the electron? We can derive it from equation I we previously used by solving it for v^2 (Fig. 1.1.20). We can then insert the term for v^2 into the equation $E_{kin} = 1/2 mv^2$, which gives $E_{kin} = e^2 / (8\pi\epsilon_0 r)$, Fig. 1.1.23.

Now let us consider the potential energy. The potential energy is the Coulomb energy between the proton and the electron in the H atom. The formula for the Coulomb energy of two particles having two opposite elementary charges is $E_{pot} = -e^2 / (4\pi\epsilon_0 r)$. Note that this energy has a negative algebraic sign because the forces between the proton and the electron are attractive (Fig. 1.1.23).

$$\begin{aligned}
 E &= E_{kin} + E_{pot} \\
 E_{kin} &= \frac{1}{2}mv^2 & E_{pot} &= -\frac{e^2}{4\pi\epsilon_0 r} \\
 \text{Equation 1: } \frac{mv^2}{r} &= \frac{e^2}{4\pi\epsilon_0 r^2} \\
 v^2 &= \frac{e^2}{4\pi\epsilon_0 r m} \\
 E_{kin} &= \frac{e^2}{8\pi\epsilon_0 r}
 \end{aligned}$$

Figure 1.1.23 Derivation of kinetic, potential, and overall energies of the H electron (part 1).

We can now add up the kinetic and the potential energy to give the overall energy which is $E = E_{kin} + E_{pot} = [e^2/(8\pi\epsilon_0 r)] - [e^2/(4\pi\epsilon_0 r)] = e^2/(\pi\epsilon_0 r) (1/8 - 1/4) = -e^2/(8\pi\epsilon_0 r)$. We can then use the term we previously calculated for r (Fig. 1.1.21) and insert it into the term for the overall energy. As a result the overall energy becomes $E = -(e^4 m)/(8\epsilon_0^2 n^2 h^2)$, Fig. 1.1.24. As we can see, the energies for the electrons in the different orbits are also only a function of the quantum number n , specifically, they are a function of $1/n^2$. Note also that the overall energy is negative. This is because the energy is a binding energy. Because of the negative algebraic sign, a higher quantum number n means a higher, because less negative energy. At very high quantum numbers n the value for E would approach zero, meaning that the binding energy for the electron would approach zero. The higher the orbit of the electron the more energy it has and the less strongly it is bound to the atom.

$$\begin{aligned}
 E &= E_{kin} + E_{pot} = \frac{e^2}{8\pi\epsilon_0 r} - \frac{e^2}{4\pi\epsilon_0 r} = \frac{e^2}{\pi\epsilon_0 r} \left(\frac{1}{8} - \frac{1}{4} \right) = -\frac{e^2}{8\pi\epsilon_0 r} \\
 \text{B/c we had calculated before: } r &= \frac{n^2 h^2 \epsilon_0}{\pi m e^2} \longrightarrow E = -\frac{e^4 m}{8\epsilon_0^2 n^2 h^2}
 \end{aligned}$$

Figure 1.1.24 Continued derivation for the energies of the H electron.

$$\begin{aligned}
 &\text{Formula for energy of electron in a specific orbit:} \\
 E &= -\frac{e^4 m}{8\epsilon_0^2 n^2 h^2} \\
 &\text{Formula for energy difference between two orbits:} \\
 \Delta E &= -\frac{e^4 m}{8\epsilon_0^2 n_{high}^2 h^2} - \left(-\frac{e^4 m}{8\epsilon_0^2 n_{low}^2 h^2} \right) = \frac{e^4 m}{8\epsilon_0^2 h^2} \left(\frac{1}{n_{low}^2} - \frac{1}{n_{high}^2} \right) \\
 &\text{Empirically Rydberg found the H spectrum:} \\
 \Delta E &= R_H \left(\frac{1}{n_{low}^2} - \frac{1}{n_{high}^2} \right) \\
 &\text{Because } \Delta E \text{ (theory)} = \Delta E \text{ (empirical):} \\
 R_H &= \frac{e^4 m}{8\epsilon_0^2 h^2}
 \end{aligned}$$

Figure 1.1.25 Comparing derived energies of the H electron with those experimentally found in the H spectrum.

Finally let us calculate energy differences between electrons in different orbits. Subtraction of the terms for the energies of two electrons in two different orbits gives $\Delta E = (e^4 m)/(8\epsilon_0^2 h^2) (1/(n_{low}^2) - 1/(n_{high}^2))$, Fig. 1.1.25. The calculated and empirically found ΔE match excellently, the empirically found Rydberg constant matches the theoretically derived constant $(e^4 m)/(8\epsilon_0^2 h^2)$. Thus experiment and theory are in accordance. Bohr's theory is able to explain the H spectra very well, and can predict both radii of electron orbits and energies. The Bohr model for the first time introduced the quantization of electron states in atoms, and in this regard it was a big step forward. However, there were still problems with Bohr's theory. It could only explain the H spectra well, but failed to explain the spectra of all other atoms. Secondly, Bohr's postulates seemed ad hoc and lacked an explanation. There was no good explanation why an electron in a quantized orbit would not emit electromagnetic radiation continuously. Thus, the Bohr model could still not be the final answer to atomic theory. In fact, it lacks to take an important property of the electron into account: The wave-particle dualism of the electron.

Definition: Problems with Bohr's Theory

1. It can explain only the H spectrum and fails to explain all the spectra of all other atoms.
2. Bohr's postulates are ad hoc. They lack an explanation.

Dr. Kai Landskron ([Lehigh University](#)). If you like this textbook, please consider to make a donation to support the author's research at Lehigh University: [Click Here to Donate](#).

This page titled [1.1: Historical Development of Atomic Theory](#) is shared under a [not declared](#) license and was authored, remixed, and/or curated by [Kai Landskron](#).

- [1.1: Historical Development of Atomic Theory](#) by [Kai Landskron](#) is licensed [CC BY 4.0](#).

1.2: The quantum-mechanical model of the atom

Wave Particle Dualism

The phenomenon of the wave-particle dualism was first discovered for electromagnetic radiation, and then extended to all other particles including the electron. It began with the investigation of the photoelectric effect by Albert Einstein (Fig. 1.2.1 and Fig. 1.2.2).

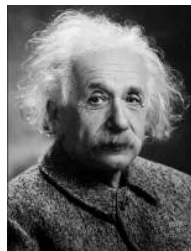


Figure 1.2.1 Albert Einstein, Nobel Prize 1921. (Attribution: Photograph by Orren Jack Turner, Princeton, N.J. Modified with Photoshop by PM_Poon and later by Dantadd. [Public domain], commons.wikimedia.org/wiki/File:lanck_1933.jpg)

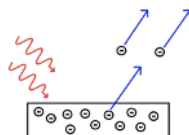


Figure 1.2.2 The photoelectric effect (Attribution: Wolfmankurd [CC BY-SA (<http://creativecommons.org/licenses/by-sa/3.0/>)], commons.wikimedia.org/wiki/File:ric_effect.svg)

The photoelectric effect occurs when a metal surface is irradiated by light. Above a certain frequency, or below a certain wavelength, light is able to eject electrons from the metal surface. The threshold frequency depends on the metal. Below the threshold frequency no electrons get ejected. Einstein investigated the maximum kinetic energy of the ejected electrons as a function of the frequency of the light. He found that there was a linear relationship. He analyzed the slope of this line and found that the slope was the Planck constant h . This would mean that electrons had an energy $E = h\nu$ minus an energy E_B that would be needed to overcome the binding energy, also called the work function, of the electron in the metal (Figure 1.2.3, left).

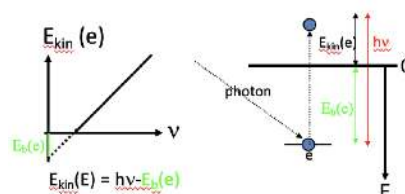


Figure 1.2.3 Maximum kinetic energy of the ejected electron as a function of the frequency of light (left), and mechanistic interpretation of the photoelectric effect (right).

The equation $E = h\nu$ was previously derived by Planck (Fig. 1.2.4) based on the assumption that energy was quantized, and now Einstein had experimentally found it again in the quest to explain the photoelectric effect. This would mean that light was quantized. The quantization would be explained by the fact that light would not only have wave but also particle properties, and these particles would be called photons. Assuming photons the photoelectric effect could be easily explained (Figure 1.2.3, right). When light hits the metal surface the photon collides with the electron. Only when the photon had an energy larger than the work function of the metal, the electron would be ejected and would have a kinetic energy equal to the difference between the energy of the photon and the binding energy. The wave-particle dualism of light, and electromagnetic radiation in general can also be mathematically derived. Because mass can be converted into electromagnetic radiation according to the equation $E = mc^2$, and the energy of electromagnetic radiation is $E = h\nu$, $mc^2 = h\nu$. We can solve the equation for ν , and then it is $\nu = mc^2/h$. With $\nu = c/\lambda$, and solved for λ , the equation becomes $\lambda = h/mc$. This equation shows the wave particle-dualism of electromagnetic radiation because it relates a wavelength to a mass. In fact, the mass of the particle associated with electromagnetic radiation, the photon, is inverse proportional to the wavelength of the electromagnetic radiation. The discovery of the wave-particle dualism of electromagnetic radiation was a radically new concept that is difficult to grasp intellectually up to this date because the human mind tends to see waves and particles to be mutually exclusive. However, it is one of the most fundamental principles of nature. As we will see later, not only electromagnetic radiation shows the wave particle dualism, but all particles including electrons.



Figure 1.2.4 Max Planck, Nobel Prize 1918 (Attribution: commons.wikimedia.org/wiki/File:lanck_1933.jpg)

Wave Particle Dualism of Massive Particles

The wave-particle dualism was originally thought to be valid for the photon only. A young French physics PhD student, Louis De Broglie had the radical idea that not only the photon, but all particles would exhibit the wave particle dualism, including the electron. Einstein's formula $\lambda = h/mv$ would just need to be slightly rearranged into $\lambda = h/mv$, whereby m would be the mass of the particle, and v would be the velocity of the particle. This idea was much disputed at the time, and Louis De Broglie's PhD thesis was almost not accepted. However, eventually the wave-particle dualism of the electron was proven by electron diffraction experiments, and Louis De Broglie was awarded the Nobel prize in 1929. Today, we believe that all particles show the wave-particle dualism, and no experiment up to this date indicates an exception.



Figure 1.2.5 Louis de Broglie 1892-1987, Nobel Prize 1929 (Attribution: Unknown commons.wikimedia.org/wiki/F...roglie_Big.jpg, PD-US expired)

Standing Waves

With the discovery of the wave-particle duality of the electron, and the observation of the quantization of electronic states in atoms led physicists focus on a field in physics in which waves are quantized. The field of standing waves (Fig. 1.2.6).

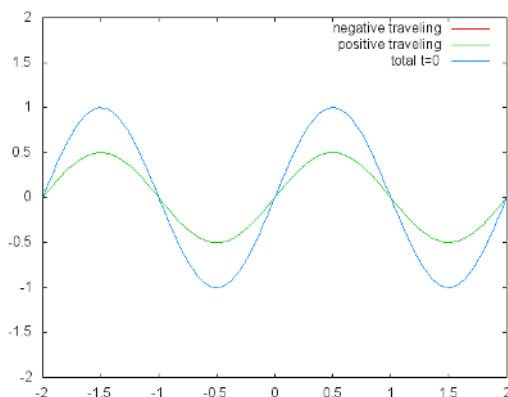


Figure 1.2.6 Two confined traveling waves (red and green) producing a standing wave (blue) due to their interference. (Attribution: Govindabalan [CC BY-SA (<https://creativecommons.org/licenses/by-sa/3.0/>)], commons.wikimedia.org/wiki/F...animation.gif)

Standing waves are a quite common phenomenon. You can for example trigger a standing waves by plucking a guitar string. This causes the guitar string to vibrate in standing waves with discrete, quantized wavelengths. The vibration with the longest wavelength is the so-called ground vibration. Its wavelength is two times the length of the guitar string. In addition, so-called higher harmonics are possible. The first harmonic has a wavelength equal to the length of the guitar string, the second one, has a wavelength equal to two-thirds of the length of the string, the third one has a wavelength equal to half of the length of the string, the fourth one has a wavelength equal to one third of the length of the guitar string, and so fourth (Fig. 1.2.7).



Figure 1.2.7 Standing waves (Attribution: Adjwilley [CC BY-SA (<https://creativecommons.org/licenses/by-sa/3.0/>)], commons.wikimedia.org/wiki/F...n_a_string.gif)

It can be easily seen that the possible wavelengths at which the string can vibrate follow the equation $\lambda = 2L/n$, whereby n is an integer, or a quantum number, and L is the length of the guitar string. Thus, we can say that the waves associated with the guitar string vibrations are quantized. Why are these waves called standing waves? This is because the positions of the crests and the troughs and the nodes do not move. They remain at the same position on the guitar string at any point in time. It should be said here that the standing nature of the waves is actually an illusion. There are actually two waves traveling in opposite direction on the guitar string, and these waves interfere with each other so that a standing wave is produced (this is illustrated in Fig. 1.2.6). When the guitar string is plucked two waves are sent into opposite direction on the guitar string toward the two opposite ends on the guitar string. Once they have reached these ends they get reflected and sent into the opposite direction until they again reach the ends of the guitar string where they get reflected again. During this process, which happens over and over again, the two waves interfere and produce the standing wave. In sum, the fact that the wave is confined within the guitar string, leads to the quantized standing waves.

Electron in a One-Dimensional Box

Let us now go from a vibrating guitar string to an electron in a one-dimensional box of length a having infinitely high walls. Inside the box the potential energy of the electron is zero, in the walls the potential energy is infinite (Figure 1.2.8).

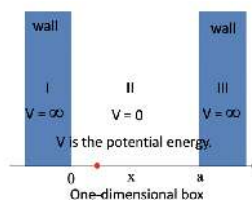


Figure 1.2.8 Electron (red dot) in a one-dimensional box with a potential energy $V=0$ within the box and infinite potential energy within the walls (the walls may be thought infinitely high).

Due to its kinetic energy the electron can travel on a line within the box until it hits the wall (Fig. 1.2.9, A). At the wall it is getting reflected and forced to travel into the opposite direction until it again hits the wall where it reverses direction again, and so forth. Consider now that the electron does not only have particle, but also wave properties. Because of that also a wave travels along the line, gets reflected at the wall, travels into opposite direction, gets reflected again and so on. These waves can interfere with it other just like the waves traveling on the guitar string to produce a standing wave. Thus, the electron in the one-dimensional box should behave like a standing wave, and this wave should be quantized (Fig. 1.2.9, B-F).

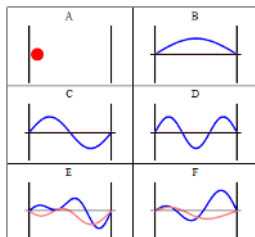


Figure 1.2.9 Electron traveling in a one-dimensional box acting as a standing wave (Attribution: Sbyrnes321 [CC0], commons.wikimedia.org/wiki/File:1Animation.gif)

How can we mathematically describe the electron in the box as a standing wave? Generally, you can describe a standing wave by a wave function. A wave function tells the amplitude of the standing wave at a particular position in the one-dimensional box. How can we find the wave function? We can start out from a differential equation that is generally valid for standing waves (Eq. 1.2.1).

$$\frac{d^2 \Psi(x)}{dx^2} = -\left(\frac{2\pi}{\lambda}\right)^2 \Psi(x)$$

Equation 1.2.1 Standing wave differential equation.

It says that the second derivative of the amplitude of the wave function at the position x is equal to $-(2\pi/\lambda)^2$ multiplied with the amplitude of the wave function at the position x . Let us now consider that the kinetic energy of the electron is $E = 1/2mv^2$ and expand the equation by m .

$$\begin{aligned} E &= \frac{1}{2}mv^2 \\ E &= (mv)^2/2m \rightarrow (mv)^2 = 2mE \rightarrow mv = [2mE]^{1/2} \rightarrow \lambda = h/[2mE]^{1/2} \\ \lambda &= h/mv \rightarrow mv = h/\lambda \end{aligned}$$

Equation 1.2.2 Solving for λ from kinetic energy equation and De Broglie equation.

Let us then solve the equation for (mv) . Now let us solve the De Broglie equation for mv and insert mv by h/λ in the previous equation. Finally, let us solve the equation by λ and we will get $\lambda = h/[2mE]^{1/2}$, Eq. 1.2.2. We can now substitute λ by $h/[2mE]^{1/2}$ in the differential equation. Slightly rearranged this equation becomes the Schrödinger equation for the electron in the one-dimensional box (Equation 1.2.3).

$$\left[\frac{d^2}{dx^2} - \frac{h^2}{8\pi^2m} \right] \Psi(x) = E_{\text{kin}} \Psi(x)$$

Equation 1.2.3 Schrödinger Equation for the electron in a 1-D box.

The Schrödinger equation is a differential equation. To get the wave function that describes the electron in the box we need to solve the differential equation. One possibility solve a differential equation is to guess its solution, and after that show that the solution is right. This is the approach we want to pursue here. A very general wave function is one that is a sum of a sinus term and a cosinus term of the coordinate x whereby we shall assume two general coefficients r and s in front of x , and two other general coefficients A and B in front of the sinus and the cosinus term, respectively.

$$\Psi = A \sin rx + B \cos sx$$

Equation 1.2.4 Generic solution to the differential equation.

We can now think of so-called boundary conditions for the wave function which will make the wave function more specific. A boundary condition is a property the wave function must have to be a sensible solution to the differential equation. We can assume a first boundary condition which assumes that at the position $x=0$ the amplitude of the wave function must be 0. This can be assumed because at these positions the "electron wave" gets reflected at the wall. That means that the wave function cannot have a cosinus term and thus B must be 0. If the wave function had a cosinus term it would not be 0 at $x=0$ because the cosinus of 0 is not 0.

$$B = 0 \quad \Psi = A \sin rx$$

Equation 1.2.5 Boundary condition 1 for the wave function

The second boundary condition is that the amplitude of the wave function is zero at $x = a$. Again, this is because the electron hits the wall at $x = a$ and reverses its direction. The sinus function is only zero at $x = a$ when ra is an integer number n times π : $ra = n\pi$. This means that r must be $n\pi/a$.

$$ra = n\pi \text{ or } r = n\pi/a$$

Equation 1.2.6 Boundary condition 2 for the wave function

Thus, the wave function must be $\Psi = A \sin(n\pi x/a)$. We can see that the wave function that describes the electron as a standing wave in the one-dimensional box is quantized because the quantum number n appears.

$$\Psi = A \sin(n\pi x/a)$$

Equation 1.2.7 The wave function considering boundary conditions 1 and 2.

Inserting the quantum numbers n into the wave functions produces all the standing waves the electron can adopt (Fig. 1.2.10). You can see that the number of nodes and the wavelengths of the waves depend on the quantum number n . For $n = 0$ there is no node and the wavelength is twice the length of the box, for $n = 2$ there is one node and the wavelength is equal to the length of the box, for $n = 3$ there are three nodes and the wavelength of the wave is $2/3$ of the lengths of the box and so forth. We can illustrate that the wave function describes the waves depicted in Figure 1.2.10 by an example. The amplitude for the wave for $n=2$ is zero in the middle of the box where $x = a/2$. If we insert $a/2$ into the equation for Ψ then $\Psi = A \sin(n\pi 2a/a) = A \sin(n\pi) = 0$ because it is the property of a sinus function to be 0 at an integer number multiple of π . We could also insert other values for x and n into the wave function, and would get the expected amplitude. This shows that the wave function correctly represents the standing waves the electron can adopt.

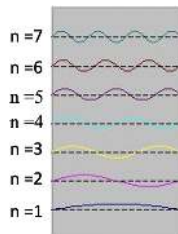


Figure 1.2.10 Standing waves corresponding to the wave function for different quantum numbers n .

We are still not quite finished with the wave function because we have not determined the parameter A in front of the sinus term. To obtain it we need to consider a third boundary condition (Eq. 1.2.8).

$$\int_0^a \Psi^2 dx = 1$$

Equation 1.2.8 Boundary condition 3.

It says that the integral of the square of the wave function over the length of the box must be equal to one. We can understand this boundary condition when we consider that the square of the wave function represents the probability to find the electron at a particular position in the box. The value the square of Ψ adopts at a position x within the box represents the probability to find the electron at this position when we consider the electron as a particle. This is called the Born interpretation of the wave function, named after the German physicist Max Born.

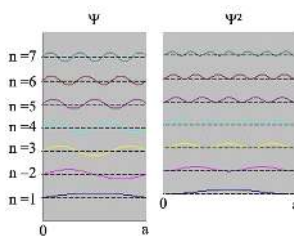


Figure 1.2.11 Standing waves corresponding to the wave function and its square.

Because the probability to find the electron anywhere in the box must be 100%, the integral of the square of the wave function over the entire box must be 100% or 1. One can show, and we omit the necessary mathematical steps for clarity here, that the boundary condition is only fulfilled when A is equal to the square root of $2/a$. The final wave function is then Ψ is equal to $\sqrt{2/a} \sin(n\pi x/a)$. The factor square root of $2/a$ is called the normalization constant of the wave function because it adjusts the amplitude of the wave function so that the probability to find the electron anywhere in the box is 100%.

The Schrödinger Equation for the H Atom

Let us now go from an electron in a 1-dimensional box to the electron in the hydrogen atom. What is similar and what is different between these two cases. A similarity is that the electron is confined in an atom just like the electron is confined in the 1D box. Thus, like in the 1-dimensional box the electron should behave like a standing wave. While in the one-dimensional box there is only one coordinate to consider, there are three coordinates to consider for the electron in the atom. This is because an atom is spherical and thus three coordinates x , y , and z are necessary to describe a position within the atom. A second major difference is that the potential energy of the electron is zero at any position within the box, while it is not zero in the hydrogen atom. This is because in an atom there are attractive Coulomb forces between the nucleus and the electron. The further away the electron is from the nucleus the higher its potential energy. This is because it takes energy to pull the electron away from the nucleus.



Figure 1.2.12 Erwin Schrödinger (1887-1961), Nobel Prize 1933 (Attribution: Nobel foundation [Public domain] commons.wikimedia.org/wiki/F...ger_(1933).jpg)

We therefore need to modify the Schrödinger equation that we used previously for the one-dimensional box the following way. Firstly, we need to expand the operator for the kinetic energy from one to three dimensions and introduce the coordinates y and z in addition to x . Secondly, we have to add an operator for the potential energy to the equation. The potential energy is the Coulomb energy between the proton and the electron. It is similar to the term we previously used for the calculation of the potential energy of the electron in the Bohr model. Instead of the radius r we use now the square root of the sum of the square of the three coordinates x , y , and z , to indicate the Coulomb energy of the electron at any position within the atom. Our wave function will now be a function of three coordinates x , y , and z . This means our wave function will now be three-dimensional and represent three-dimensional standing

waves. Three-dimensional waves are harder to imagine compared to 1-dimensional ones, but have the same properties, which are that the position of the crests, troughs, and nodes does not move.

$$\left[\left(\frac{d^2}{dx^2} + \frac{d^2}{dy^2} + \frac{d^2}{dz^2} \right) \frac{-\hbar^2}{8\pi^2m} + \frac{e^2}{4\pi\epsilon_0 \sqrt{x^2 + y^2 + z^2}} \right] \Psi(x,y,z) = E \Psi(x,y,z)$$

Operator for kinetic energy + Operator for potential energy
(Coulomb energy)

Equation 1.2.9 Schrödinger equation for the H atom.

In order to get the wave functions for the electron we need to solve the Schrödinger equation for the hydrogen atom. To do so, we need to think about the boundary conditions for the wave function. One condition is that the square of the wave function approaches zero when we go very far from the nucleus, and r , the distance from the nucleus approaches infinite. Secondly, like in the one-dimensional box, the integral over the square of the wave function must be one. This is because the probability to find the electron somewhere in the atom must be 100%. Thirdly, it would be sensible to assume that the wave function must be continuously differentiable and single valued (Fig. 1.2.13).

Boundary conditions for Ψ :
 $\Psi^2 = 0$ for $r \rightarrow \infty$
 $\int \Psi^2 dr = 1$
 -continuously differentiable and single valued

Figure 1.2.13 Boundary conditions for the wave function of the electron in the hydrogen atom.

The Spherical Polar Coordinate System

The mathematical process to solve the Schrödinger equation is beyond the scope of this course and you are referred to Physical Chemistry classes and textbooks for the details. We shall only provide an brief outline of the process here. It is mathematically simpler to solve the Schrödinger equation in spherical polar coordinates instead of cartesian coordinates. Therefore, we obtain the solutions of the Schrödinger equation, the wavefunctions, in polar coordinates. The position of a point is specified by three numbers: the *radial distance* of that point from a fixed origin, its *polar angle* measured from a fixed zenith direction, and the *azimuthal angle* of its orthogonal projection on a reference plane that passes through the origin and is orthogonal to the zenith, measured from a fixed reference direction on that plane (Fig. 1.2.14).

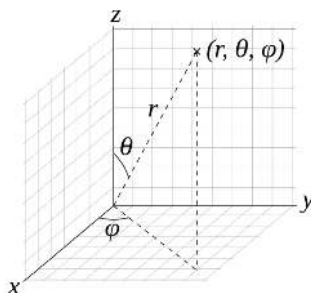


Figure 1.2.14 Illustration of the spherical polar coordinate system.

Solutions of the Schrödinger Equation

The wavefunction is then a function of r , θ and ϕ , in particular it is a product of a radial wave function which is function of r , a colatitude wave function, which is a function of θ , and an azimuthal wave function which is a function of ϕ (Eq. 1.2.10). You can see the explicit forms of the radial, colatitude, and azimuthal functions in Fig. 1.2.15.

$$\Psi(r, \theta, \phi) = R(r) \times \Theta(\theta) \times \Phi(\phi)$$

Equation 1.2.10 The wave function for the electron in the H atom as a function of r , θ and ϕ

Radial wave function	→	$R_{n,l} = r^l L_{n,l} e^{-r/na_0}$
Colatitude wave function	→	$\Theta(\theta) = (-1)^m \sqrt{\frac{2l+1}{2\pi} \frac{(l-m)!}{(l+m)!}} P_l^m(\cos \theta)$
Azimuthal equation	→	$\Phi(\phi) = \frac{1}{\sqrt{2\pi}} e^{im\phi}$

Figure 1.2.15 Explicit forms of the radial, colatitude and azimuthal functions

We do not need to understand all details here, but need to realize that these wavefunctions are functions of quantum numbers. In contrast to the electron in the one-dimensional box not only one, but three quantum numbers need to be considered. Beyond the quantum number n , which is called the principal quantum number, there is also a so-called orbital quantum number l , and a magnetic quantum number m . The quantum number n only occurs in the radial wave function, the quantum number l only occurs in the radial and colatitude function, and the quantum number m only occurs in the colatitude and the azimuthal part of the wave function. The values the orbital quantum number l can adopt depends on n , it can be between 0 and $n-1$ for a given quantum number n . The magnetic quantum number m depends on the quantum number l , and can run from $-l$ to $+l$. So, for example when n is equal to 2, l can vary between 0 and 1, and for l equal to 2, m can adopt any value between -2 and 2, namely -2, -1, 0, +1, and +2 (Fig. 1.2.16).

Principal quantum number: n
 Orbital quantum number: l ($l = 0, \dots, n-1$)
 Magnetic quantum number m ($m = -l, \dots, +l$)

Figure 1.2.16 The quantum numbers of the wave function of the H atom.

These wave functions have a particular name in chemistry. They are called orbitals. One can understand an orbital as the three-dimensional wave function that describes the electron in an atom as a standing wave. Thus, an orbital is a state the electron can adopt, and when we say that an electron is in a particular orbital we mean that the electron is in a particular state.

The spin quantum number s

Within an orbital an electron can adopt two different spins described by the spin quantum number s . This quantum number is not a result of the Schrödinger equation, but was found experimentally. The spin quantum number s can adopt two values: $+1/2$ and $-1/2$. We say an electron is spin up when $s = +1/2$, and spin down when $s = -1/2$.

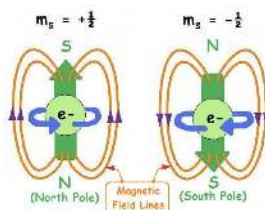


Figure 1.2.17 Electron spin (Attribution: Chemlibretexts.org chem.libretexts.org/@api/deki...jpg?revision=1, CC BY-NC-SA 3.0)

The spin quantum number is understood most easily when you view the electron as a particle rotating around its own axis. A counter-clockwise rotation would be associated with $s = +1/2$, a clockwise rotation would be associated with $s = -1/2$. The rotation produces a magnetic field with the direction of the field lines depending on the direction of rotation.

The 1s Orbital

Let us now insert the quantum numbers into the general form of the wave function for the electron in the hydrogen atom, and calculate their explicit mathematical forms. Let us start with the smallest possible numbers. When we insert $n=1$, $l=0$, and $m=0$ we get the wave function for the 1s orbital (Eq. 1.2.11).

$$\Psi_{1s} = \frac{\Phi(\phi)}{\sqrt{2\pi}} \times \frac{\Theta(\theta)}{\sqrt{2}} \times \frac{R(r)}{a_0^{3/2}} e^{-r/a_0}$$

Equation 1.2.11 Wave function for the 1s orbital

You can see that the colatitude and azimuthal parts of the wave functions, together also called the angular part of the wave function, are simple numbers, namely one over the square root of two pi, and one over the square root of two respectively. The angles theta and phi do not appear and that means that the wave function is angle-independent. This implies that the orbital has a spherical shape.

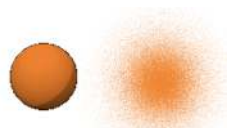


Figure 1.2.18 Common depictions of the 1s orbital

The radial part of the wave function has an exponential term e to the power of $-r$ over a_0 . The number e is the Euler number, that is the base of the natural logarithm. It is approximately equal to 2.71828. a_0 is the Bohr radius, which is the distance of the electron from the proton for the electron on its first orbit according to the Bohr atom model. We calculated it previously, and it was $5.29 \times 10^{-11}\text{m}$. r is the distance from the nucleus. This exponential term has a constant in front of it which is 1 over the Bohr radius to the power of three over two. Because the radial part of the wave function is e to the power of $-r$, the amplitude of the radial part of the wave function declines exponentially with increasing distance r from the nucleus. This is true for the entire wave function because the angular parts of the wave function are just simple numbers. You can see the exponential decline of the amplitude of the wave function as a function of the radius r in units of the Bohr radius a_0 in the graph below (Fig. 1.2.19).

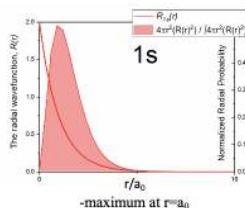


Figure 1.2.19 Amplitude of the radial wave function R of the 1s orbital as a function of the radius r in units of the Bohr radius a_0 , and radial probability function (Attribution: Chemlibretexts chem.libretexts.org/@api/deki/files/205599/clipboard_e63775e352caeb1d434fe16aa3fa22d21.png?revision=1, <http://creativecommons.org/licenses/by-nc-sa/3.0/us/>)

This means that the 1s orbital must be a spherical orbital with its amplitude exponentially declining with increasing distance from the nucleus. This can be graphically depicted via a density of points around the nucleus (Fig. 1.2.18). A higher density of points means higher amplitude and a lower density of points means smaller amplitude. The fact that the amplitude is the highest closest to the nucleus means that the probability to find the electron in a particular point in space is the highest closest to the nucleus. This is because the square of the wave function for the 1s orbital represents the probability to find the electron at a particular point in space. However, the probability to find the electron at a certain distance from the nucleus is not the highest closest to the nucleus. This is because the probability to find the electron at a certain distance from the nucleus is the probability to find the electron at a particular point in space times the number of points in space that have that particular distance. The number of points in space that have a particular distance r from the nucleus is related to the surface of a sphere with the radius r . The surface of a sphere A is given by the formula $A = 4\pi r^2$. Thus, the probability to find the electron at a particular distance r from the nucleus is equal to the surface of a sphere with the radius r times the radial wave function square. This defines the radial probability function $R_P = 4\pi r^2 R^2$.

The graph for the radial probability function is depicted in Figure 1.2.19. You can see that it is zero when $r=0$ which is due to the r^2 term which becomes zero at $r = 0$. At small r values the $4\pi r^2$ term dominates the overall function and the probability to find the electron increases with r . However, because the exponential term of the radial function becomes more and more significant at larger r values, the radial probability function goes through a maximum and then declines. At distances r approaching infinite the function approaches zero. The maximum of the curve represents the distance at which it is most likely to find the electron. Interestingly, it is equal to the Bohr radius r . That means that Bohr was not so wrong after all. However, he was wrong in the sense that he viewed the electron as a classical particle being only at that radius. Instead, the electron, due to its wave properties, it is not only at the Bohr radius r_0 , but most likely we can find it there.

The result that the probability to find the electron at a specific point in space is the greatest at the nucleus does away with the question of why the H atom does not collapse. Because the amplitude of the wave function is the greatest closest to the nucleus we can actually say in a way that the atom is collapsed. Nonetheless, the atom is mostly empty space because of the delocalization of the wave function that describes the state of the electron. The delocalization of the electron due to its wave properties also makes the question of the size of an orbital non-trivial. The wave function approaches zero only for r approaching infinite, but never becomes zero. This means that to account for all electron density the orbital would be infinitely large. This however would not be a sensible definition. The commonly accepted definition is that the size of an orbital is defined by the space that contains 90% of its electron density. We

can depict orbitals according to this definition. For the case of the 1s orbital the radius of the sphere that represents the 1s orbital is chosen so that the probability to find the electron within a sphere of that radius is 90%.

The 2s Orbital

Inserting $n=2$, $l=0$, and $m=0$ into the general wave function gives us the wave function for the 2s orbital (Eq. 1.2.12).

$$\Psi_{2s} = \frac{1}{\sqrt{2\pi}} \times \frac{1}{\sqrt{2}} \times \frac{1}{2\sqrt{2} a_0^{3/2}} \left[2 - \frac{r}{a_0} \right] e^{-r/2a_0}$$

Equation 1.2.12 Wave function for the 2s orbital

Like in the case of the 1s orbital, the angular parts of the wave function are only simple numbers. Thus, the 2s orbital is also a spherical orbital. Like in the 1s orbital, the radial part of the wave function has an exponential term of the type e^{-r} and a simple coefficient. However, there is an additional term $2-r/a_0$. Due to the exponential term, the amplitude of the wave function declines exponentially with r . However, because of the term $2-r/a_0$ the wave function becomes negative, and then approaches 0. The radius at which the wave function changes its algebraic sign is called a spherical node. The node is spherical because it describes the surface is a sphere. On the surface of this sphere the amplitude of the wave function is zero. The radius at which the wave function changes its algebraic sign is $2a_0$. This becomes clear when we consider that the term $2-r/a_0$ becomes zero when $r=2a_0$ because $2-2a_0/a_0=0$. When this term is zero then the entire wave function becomes zero. We can illustrate the change of the algebraic sign in the 2s orbital using colors as shown below (Fig. 1.2.20).

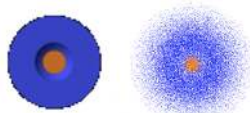


Figure 1.2.20 Common depictions of the 2s orbital.

The orange sphere indicates the space in which the amplitude of the 2s orbital is positive, and the blue area indicates where it is negative. The interface area where the color changes represents the node. By multiplying the square of the radial wave function R with $4\pi r^2$ we obtain the radial probability function for the 2s orbital (Fig. 1.2.21).

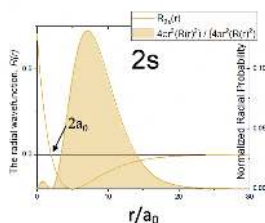


Figure 1.2.21 Amplitude of the wave function of the 2s orbital as a function of the radius r in units of the Bohr radius a_0 as well as the radial probability function (Attribution: Chemlibretexts <https://chem.libretexts.org/@api/dek...png?revision=1, http://creativecommons.org/licenses/by-nc-sa/3.0/us/>)

You can see that similar to the 1s orbital, the probability to find the electron on a radius closest to the nucleus is 0. However, unlike the 1s orbital there are now two maxima instead of only one. This second maximum is imposed by the spherical node of the 2s orbital. The second maximum is larger than the first one. Generally, the probability to find the electron further away from the nucleus is greater compared to the 1s orbital. That means that the 2s orbital is larger than the 1s orbital. This reflects the general trend that orbitals with a higher quantum number n tend to be larger and the probability to find the electron further away from the nucleus is greater.

The 3s Orbital

Let us look at one more s orbital, the 3s orbital (Eq. 1.2.13).

$$\Psi_{3s} = \frac{1}{\sqrt{2\pi}} \times \frac{1}{\sqrt{2}} \times \frac{2}{81\sqrt{3} a_0^{3/2}} \left[27 - 18 \frac{r}{a_0} + 2 \frac{r^2}{a_0^2} \right] e^{-r/3a_0}$$

Equation 1.2.13 Wave function for the 3s orbital.

The angular parts of the wave function for the 3s orbital are again simple numbers, which implies that the orbital is again spherical. Generally all s orbitals are spherical.

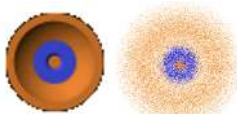


Figure 1.2.22 Common depictions of the 3s orbital

The radial part of the wave function can be divided into the parts. The first part is a simple number, then there is the term $27-18r/a_0 + 2r^2/a_0^2$, and the exponential term $e^{-r/3a_0}$. Because of the exponential part the amplitude of the wave function declines exponentially with increasing r . However, then the wave function becomes negative, goes through a minimum, then becomes positive again, goes through a small maximum, and finally approaches zero. The fact that the wave function changes its algebraic sign two times means that the 3s orbital has two spherical nodes. These spherical nodes are due to the second term which is a square function of the type $ax^2 + bx + c$, whereby in this case $a = 2/a_0^2$, $b = -18/a_0$ and $c=27$. When this term becomes zero the entire wave function becomes zero. Square functions have two solutions according to the formula $x = (-b \pm \sqrt{b^2 - 4ac})/2a$. The solutions give the radii for the two spherical nodes of the 3s orbital. We omit the exact calculation of the radii for brevity and clarity reasons here. You may calculate them as a homework.

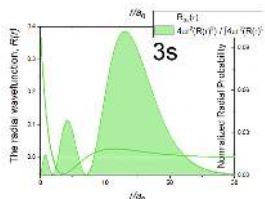


Figure 1.2.23 Amplitude of the wave function of the 3s orbital as a function of the radius r in units of the Bohr radius a_0 as well as radial probability function. (Attribution: Chemlibretexts <https://chem.libretexts.org/@api/dek...png?revision=1, http://creativecommons.org/licenses/by-nc-sa/3.0/us/>)

Because of the two spherical nodes, the radial probability function R_p has three maxima. A very small one close to the nucleus, a larger one further away from the nucleus, and the largest one at even greater distance from the nucleus. Overall, the 3s orbital is larger than the 2s orbital confirming the general trend that s orbitals increase in size with increasing quantum number n .

The $2p_z$ Orbital

Now let us look at orbitals in which the orbital quantum number l is larger than 0. When l is equal to 1, then n must be at least 2. An orbital with $n=2$ and $l=1$ is a 2p orbital. Overall, three 2p orbitals must exist because for $l=1$, the magnetic quantum number m can adopt three values: -1, 0, and +1. The orbital with $m=0$ is called the $2p_z$ orbital. Let us look at the mathematical form of this wave function (Eq. 1.2.14).

$$\Psi_{2p_z} = \frac{1}{\sqrt{2\pi}} \times \frac{\sqrt{6}}{2} \cos \theta \times \frac{1}{2\sqrt{6} a_0^{3/2}} \frac{r}{a_0} e^{-r/2a_0}$$

Equation 1.2.14 Wave function for the $2p_z$ orbital

Analyzing the angular parts of the wave function shows that while the azimuthal part is still a simple number, the colatitude part is not, it is a function of θ . This means that the amplitude of the wave function is now angle dependent, which implies that the orbital cannot be spherical. Instead the 2p orbital has the shape of a dumbbell which is oriented along the z-axis. One lobe is above the xy plane, and the other lobe is below the xy plane. These two different lobes have a different algebraic sign, indicated by different colors (Fig. 1.2.24).

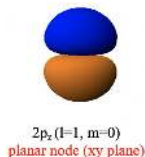


Figure 1.2.24 Common depiction for the $2p_z$ orbital

Within the xy plane the amplitude of the wave function is zero. The xy plane represents a so-called planar node. A planar node is a type of an angular node. The name angular node is because it is due to the fact that the angular part of the wave function is not a simple number, but a function of an angle, in this case θ . We can show that the angular part of the wave function produces the planar node in the xy plane when we convert it from spherical into Cartesian coordinates (Eq. 1.2.15).

$$\Theta\Phi(x, y, z) = \frac{1}{2} \sqrt{\frac{3z}{\pi r}} \text{ is 0 for } z = 0$$

Equation 1.2.15 Angular part of the wave function of $2p_z$ in cartesian coordinates. The function becomes zero when $z=0$.

We can see that the angular part of the wave function is only a function of z . We can also see, that the angular part of the wave function becomes zero when $z=0$. When the angular part of the wave function becomes zero, then the wave function of the entire orbital becomes zero. z is zero only in the xy plane, and this explains why the xy plane is a planar node.

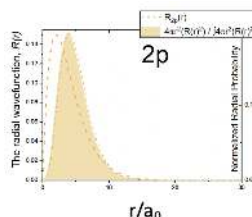


Figure 1.2.25 Amplitude of the wave function of the $2p_z$ orbital as a function of the radius r in units of the Bohr radius a_0 as well as radial probability function. (Attribution: Chemlibretexts
<https://chem.libretexts.org/@api/dek...png?revision=1>, <http://creativecommons.org/licenses/by-nc-sa/3.0/us/>)

Now let us look at the radial part of the wave function. It is made of three terms: a simple constant, the term r/a_0 , and the exponential term $e^{-r/2a_0}$. Because of the term r/a_0 the radial part of the wave function is zero at $r=0$. This means that it is very unlikely to find the electron at a point in space directly at the nucleus. This behavior is opposite to s orbitals. Because the r/a_0 term dominates the behavior of the wave function at small r value, the amplitude first increases. However, at larger r values the exponential term becomes more and more important which forces the amplitude of the radial function through a maximum, after which it declines and approaches zero. The radial wave function never changes algebraic sign. Therefore, the $2p_z$ orbital does not have a spherical node. The radial probability function has a similar shape compared to the radial function. It is zero at $r=0$ and goes through a maximum before it declines again and approaches zero (Figure 1.2.25).

The $2p_x$ and $2p_y$ Orbital

The $2p_x$ and the $2p_y$ orbitals are obtained when the quantum numbers m are 1, and -1, respectively.

$$\Psi_{2p_x} = \frac{1}{\sqrt{2\pi}} e^{\pm i\phi} \times \frac{\sqrt{2}}{2} \sin \theta \times \frac{1}{2\sqrt{6} a_0^{3/2}} \frac{r}{a_0} e^{-r/2a_0}$$

Equation 1.2.16 Wave function for the $2p_x$ and $2p_y$ orbital

They have the same shape as the $2p_z$ orbital, but their orientation is different. The dumbbell of the $2p_x$ orbital is oriented along the x-axis, and that of the $2p_y$ orbital is oriented along the y-axis.

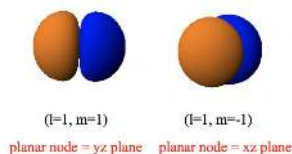


Figure 1.2.26 Common depictions for the $2p_x$ and $2p_y$ orbitals, respectively

This is because the $2p_x$ orbital has a planar node in the yz plane, and the $2p_y$ orbital has a planar node in the xz plane. The planar nodes are due to the angular part of the wave functions of these orbitals.

$$\Theta\Phi(x, y, z) = \frac{1}{2} \sqrt{\frac{3x}{\pi}} \text{ is 0 for } x = 0$$

$$\Theta\Phi(x, y, z) = \frac{1}{2} \sqrt{\frac{3y}{\pi}} \text{ is 0 for } y = 0$$

Equation 1.2.17 Angular part of the wave function of $2p_x$ and $2p_y$, respectively. The wave functions become 0 for $x=0$ and $y=0$, respectively.

As one can see, the angular parts of the wave function contain both the angles θ and Φ . When converted into Cartesian coordinates (Eq. 1.2.17) one can see that the angular function of the $2p_x$ orbital is only a function of x , and the angular function of the $2p_y$ orbital is only a function of y . The functions become zero for $x=0$ and $y=0$, respectively. x is zero only in the yz plane, and y is zero only in the xz plane, hence these nodes are planar nodes. The radial wave function is exactly the same as for the $2p_z$ orbital.

The 3p Orbitals

Now let us look at the 3p orbitals. In a 3p orbital the quantum number $n=3$ and $l=1$. Like for the case of the 2p orbitals there are three p orbitals because the magnetic quantum number m can adopt the values -1, 0, and +1. The wave function of the $3p_z$ orbital is shown below (Eq. 1.2.18).

$$\Psi_{3p_z} = \frac{1}{\sqrt{2\pi}} \frac{\sqrt{6}}{2} \cos \theta \frac{4}{81\sqrt{6} a_0^{3/2}} \left[6 - \frac{r}{a_0} \right] \frac{r}{a_0} e^{-r/3a_0}$$

Equation 1.2.18 Wave function for the $3p_z$ orbital

One can see that the angular part of the wave function is exactly the same as for the $2p_z$ orbital. This means that like in the $2p_z$ orbital there must be a planar node in the xy plane. When you look at the radial part of the wave function you can see that it is composed of four terms, a simple constant, a term $6-r/a_0$, the term r/a_0 , and the exponential term $e^{-r/3a_0}$. Because of the term r/a_0 the wave function becomes zero at $r=0$. Because of the term $6-r/a_0$ the wave function has a spherical node. This is because when $6-r/a_0=0$ the entire wave function becomes 0. We can determine the radius at which the wave function becomes zero by solving the equation $6-r/a_0=0$ for r , which gives $r=6a_0$. Overall, the radial function is zero at $r=0$, goes through a maximum, changes its algebraic sign at the spherical node at $r=6a_0$, has a minimum, and then approaches zero for very large distances r (Fig. 1.2.28).

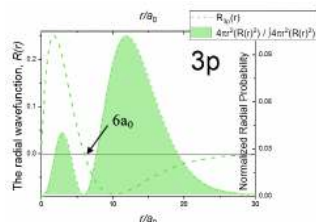


Figure 1.2.28 Amplitude of the wave function of the $3p_z$ orbital as a function of the radius r in units of the Bohr radius a_0 as well as radial probability function

The radial probability function has two maxima, a small one close to the nucleus, and a larger one further away from the nucleus. Overall the radial probability function is further away from the nucleus compared to the 2p orbitals, which implies that the 3p orbital is larger.

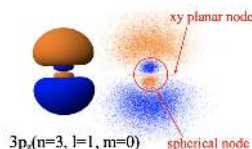


Figure 1.2.29 Common depiction for the $3p_z$ orbital

The overall shape of the $3p_z$ orbital is determined by both the spherical and the planar node (Fig. 1.2.29). Due to the planar node in the xy plane it has a dumbbell shape and is oriented in z direction. However, because of the spherical node the algebraic sign of the wave function changes within the two lobes of the dumbbell. Finally let us consider the $3p_x$ and the $3p_y$ orbitals. They would have the same shape as the $3p_z$ orbital, but they would be oriented in the x and the y direction respectively. We could also think about what would happen to the p orbitals if we increased the quantum number n further. In this case the dumbbell shape would remain, but additional spherical nodes would be introduced. A 4p orbital would have two spherical nodes, a 5p orbital three spherical nodes, and so on. The size of these p orbitals would also increase with the quantum number n .

The $3d_{z^2}$ Orbital

Now let us consider the orbitals with the orbital quantum number $l=2$. These orbitals are so-called d-orbitals. Because the principal quantum number n always has to be at least one integer number larger than the quantum l , the d orbitals with the smallest quantum number n are the 3d orbitals. For each quantum number n there must be five d orbitals because for $l=2$ the magnetic quantum number m can adopt the values -2, -1, 0, +1, and +2. Let us first look at the 3d orbital with $m=0$. This orbital is called the $3d_{z^2}$ orbital. Its wave function is shown below (Eq. 1.2.19).

$$\Psi_{3d_{z^2}} = \frac{1}{4} \sqrt{\frac{5}{\pi}} (3 \cos^2 \theta - 1) \frac{1}{81\sqrt{15} a_0^3} \frac{r^2}{a_0^2} e^{-r/3a_0}$$

Equation 1.2.19 Wave function for the $3d_{z^2}$ orbital

You can see that the radial function is a product of three terms: a constant, a term r^2/a_0^2 , and the exponential term $e^{-r/3a_0}$. The exponential term is the same as for the 3p orbitals. Because of the term r^2/a_0^2 the amplitude of the wave function is 0 at $r=0$. The amplitude then increases with r , goes through a maximum, and then approaches zero. This means that the wave function never changes its algebraic sign, and therefore it does not have spherical nodes. The radial probability function has a similar shape compared to the radial function (Figure 1.2.30). The electron probability is somewhat further away from the nucleus compared to the 3p orbitals, meaning that the 3d orbital is somewhat larger than the 3p orbital.

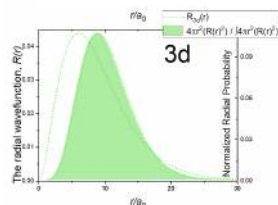


Figure 1.2.30 Amplitude of the wave function of the $3d_{z^2}$ orbital as a function of the radius r in units of the Bohr radius a_0 as well as radial probability function

You can see that the angular part of the wave function is a function of θ which means that the orbital will be non-spherical and there will be at least one angular node. In this case the angular nodes are two conical nodes. These nodes describe two cones that give the orbital its characteristic shape. It is a doughnut ring in the xy plane around a dumbbell pointing into z direction. The dumbbell and the doughnut have different algebraic signs indicated by different colors (Figure 1.2.31). Note that in contrast to the p orbitals the two lobes of the dumbbell have the same algebraic sign.

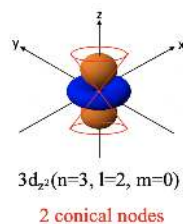


Figure 1.2.31 Common depiction for the $3d_{z^2}$ orbital with conical nodes (red).

We can understand that the $3d_{z^2}$ orbital has two conical nodes when we convert the angular part of the wave function into Cartesian coordinates (Eq. 1.2.20).

$$\Theta\Phi(x, y, z) = \frac{1}{4} \sqrt{\frac{5}{\pi}} \frac{(2z^2 - x^2 - y^2)}{r^2}$$

Equation 1.2.20 Angular part of the wave function of the $3d_{z^2}$ orbital in cartesian coordinates.

You can see that it is a function of $2z^2 - x^2 - y^2$. This is the mathematical form of cones. The angular part of the wave function becomes 0 when $2z^2 = x^2 + y^2$ which is true on the surface of two cones, one above the xy plane and one below the xy plane. The name of the $3d_{z^2}$ orbital is because it is a function of z^2 .

The $3d_{x^2-y^2}$ Orbital

Let us next look at the $3d_{x^2-y^2}$ orbital (Eq. 1.2.21).

$$\Psi_{3d_{x^2-y^2}} = \frac{1}{4} \sqrt{\frac{15}{\pi}} \sin^2 \theta \cos^2 \phi \frac{R(r)}{81\sqrt{15} \frac{1}{a_0^3} \left[\frac{2}{a_0} \right]^{3/2} \frac{r^2}{a_0^2} e^{-\frac{r}{a_0}}}$$

Equation 1.2.21 Wave function for the $3d_{x^2-y^2}$ orbital

The radial part of the wave function is the same as the one for the $3d_{z^2}$ orbital. The angular part of the wave function is both a function of θ and Φ and thus there are angular nodes to expect. We can find them again when we convert the angular part of the wave function from spherical to Cartesian coordinates.

$$\Theta\Phi(x, y, z) = \frac{1}{4} \sqrt{\frac{15}{\pi}} \frac{(x^2 - y^2)}{r^2}$$

Equation 1.2.22 Angular part of the wave function of the $3d_{x^2-y^2}$ orbital in cartesian coordinates.

You can see that the angular function is a function of $x^2 - y^2$. The function becomes zero when $x^2 - y^2 = 0$. This is the case when $x=y$ or $x=-y$. X is equal to y on two lines that bisect the 90 degree angle between the x and the y axis. The coordinate z is completely variable. Thus, overall, the wave function becomes zero on two planes that stand perpendicular to the xy plane and bisect the 90 degree angle between the x and the y axis. These two planes are therefore the two planar nodes of the orbital.

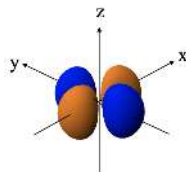


Figure 1.2.32 Common depiction of the $3d_{x^2-y^2}$ orbital

Because of the two planar nodes the orbital has four lobes lying on the x and the y axis, respectively. Lobes lying on the x -axis have the opposite algebraic sign compared to those lying on the y -axis. The name of the dx^2-y^2 orbital is because it is a function of $x^2 - y^2$.

The $3d_{xy}$, $3d_{yz}$ and $3d_{xz}$ orbitals

The remaining d orbitals are the $3d_{yz}$, the $3d_{xz}$, and the $3d_{xy}$. The radial function of these orbitals is just like the ones we previously discussed. The angular part of the wave functions is shown here directly in Cartesian coordinates. The d_{yz} orbital is a function of yz . The function becomes zero when either $y=0$ or $z=0$. y is zero in the xz plane, and z is zero in the xy plane. Therefore these planes are planar nodes of the orbital. Because of these nodes, the orbital has four lobes which lie in the yz plane whereby the lobes are in between the y and the z axes. Note that adjacent lobes have opposite algebraic sign, and opposite lobes have the same algebraic sign.

$$3d_{yz} \quad \Theta\Phi(x, y, z) = \frac{1}{2} \sqrt{\frac{15}{\pi}} \frac{yz}{r^2}$$

Figure 1.2.33 Angular wave function for the $3d_{yz}$ orbital, and the common depiction for this orbital

The $3d_{xz}$ orbital is a function of xz , and thus the function becomes zero with $x=0$ or $z=0$. This defines that yz and the xy planes as the two planar nodes. The orbital has the same shape as the $3d_{yz}$, except that the four lobes lie within the xz plane instead of the yz plane.

$$3d_{xz} \quad \Theta\Phi(x, y, z) = \frac{1}{2} \sqrt{\frac{15}{\pi}} \frac{xz}{r^2}$$

Figure 1.2.34 Angular Wave function for the $3d_{xz}$ orbital, and the common depiction for this orbital

The $3d_{xy}$ -orbital is a function of xy , and thus its wave function becomes zero at $x=0$ or $y=0$. Therefore, the xz and the yz planes are the planar nodes and the orbital has four lobes in the xy plane in between the x and the y axis. Note that the $3d_{xy}$ and the $3d_{x^2-y^2}$ look very similar but are different. The difference is that the $3d_{x^2-y^2}$ orbital has its lobes on the x and the y axis, while the $3d_{xy}$ orbital has the lobes in between the x and the y axis. One can also say that the $3d_{x^2-y^2}$ orbital is rotated by 45 degree around z with respect to the $3d_{xy}$ orbital.

$$3d_{xy} \quad \Theta\Phi(x, y, z) = \frac{1}{4} \sqrt{\frac{15}{\pi}} \frac{xy}{r^2}$$

Figure 1.2.35 Angular wave function for the $3d_{xy}$ orbital, and the common depiction for this orbital

Rules for Angular and Spherical Nodes

Of course, we could now discuss the size and shape of many other orbitals such as the $4d$ orbitals or the $4f$ orbitals, but this will be beyond scope. Instead, let us think about if there are general rules that allow the prediction of the number of spherical and angular nodes in an orbital. The number of radial nodes is always equal to $n-1$, the number of angular nodes is equal to l , and the total number of nodes is always $n-1$. This means that the number of nodes increases with the quantum number n . If we increase the quantum number l for a given quantum number n , then we replace spherical nodes by angular nodes. This is summarized in Figure 1.2.36.

- $R(r) = 0 \rightarrow$ spherical node (spherical nodal surface)
- Number of radial nodes is $n - l - 1$.
- $\Phi(\phi) = 0 \rightarrow$ angular node
(This can be a planar or conical nodal surface)
- Number of spherical nodes is equal to the orbital quantum number l .
- Total number of nodes is always $n-1$

Figure 1.2.36 Rules for angular and spherical nodes

The Orbital Energies for the H Atom

The Schrödinger equation does not only allow to calculate the orbitals of the hydrogen atom, but also their energies. The energies of Bohr and Schrödinger model match: $E = \text{constant}/n^2$. This means that orbitals with the same quantum number n have the same energies. The energies are **not** a function of the quantum numbers l and m . The energies are negative because they are binding energies. In other words: Adding an electron to a proton is an exothermic process. The binding energy for the energy increases as the orbital energy decreases (Fig. 1.2.37).

shell	n	orbitals			
		s $l=0$	p $l=1$	d $l=2$	f $l=3$
N	4	$\square$ 4s	$\square \square$ 4p	$\square \square \square$ 4d	$\square \square \square \square$ 4f
M	3	$\square$ 3s	$\square \square$ 3p	$\square \square \square$ 3d	
L	2	$\square$ 2s	$\square \square$ 2p		
K	1	$\square$ 1s			

Figure 1.2.37 Orbital energies from the 1s orbital to 4f orbital. K, L, M, N indicate the shells associated with a quantum number n .

For the 1s orbital of H the energy is -13.6 eV , the energy of the 2s and the 2p orbitals are $1/4$ of that, the energy of the 3s, 3p, and 3d orbitals are $1/9$ of the energy of the 1s orbital and so forth. Note at $1/4$ and $1/9$ th of -13.6 eV is more than -13.6 eV due to the negative algebraic sign. The electron volt is a unit of energy. It is the amount of kinetic energy gained by a single unbound electron when it passes through an electrostatic potential difference of one volt, in vacuum.

$$1 \text{ eV} = 1.602176 \times 10^{-19} \text{ J}$$

Equation 1.2.23 The electron volt to joule unit conversion.

In other words, it is equal to one volt (1 volt = 1 joule per coulomb) times the charge of a single electron (in coulombs). It is a very small unit of energy which is practical for orbital energy calculations because the orbital energies are very small (Eq. 1.2.23).

Multi-electron Atoms

Thus far we have only considered the orbitals for the hydrogen atom which contains only one electron. Can we also solve the Schrödinger equation for atoms that have more than one electron and get the exact energies of the orbitals? The answer is no, this mathematically not possible. The process is just already too complex even for only two electrons. The Schrödinger equation can only be solved for one electron systems. Therefore, the description of the atomic structure of all other atoms must work with approximations. Let us first consider the He atom. It has only one more electron than hydrogen. It is a useful approach to approximate multi-electron atoms as one-electron systems first, and then approximate the electron-electron interactions. Electron energies in an atom with more than one proton should follow the equation $E_n = -Z^2 \times 13.6 \text{ eV}/n^2$, whereby Z is the number of protons.

$$E_n = - \frac{Z^2 \cdot 13.6 \text{ eV}}{n^2}$$

Equation 1.2.24 The orbital energies for atoms with more than one proton

The binding energy of the electron increases proportionally to the square of the number of protons because the attractive Coulomb forces that act on the negatively charged electrons increases with the number of positively charged protons in the nucleus. One can experimentally measure the orbital energies via ionization energies. The energy required to remove an electron in a particular orbital from the atom is equal to the binding energy for the electron in that orbital. Therefore the ionization energy $IE = -E_n$ (E_n = orbital energy). According to the Schrödinger model the orbital energy for a helium electron in a 1s orbital should be $E_{1s} = -(2^2 \times 13.6 \text{ eV})/1^1 = -54.4 \text{ eV}$ (Eq. 1.2.25).

$$E_{1s} = -\frac{2^2 \times 13.6 \text{ eV}}{1^1} = -54.4 \text{ eV}$$

Equation 1.2.25 Energy for a helium electron in a 1s orbital.

However, the experimentally measured ionization energy of the electron is +24.6 eV, which means that the real orbital energy is -24.6 eV, and not -54.4 eV. On the other hand, the ionization of a He^+ ion is +54.4 eV which is exactly what we would expect. We can explain this phenomenon by the fact that the Schrödinger model works for a single electron only and must neglect electron interactions. In a He^+ ion there is only one electron, therefore the Schrödinger model correctly predicts the energy of the electron. However, in a helium atom there are two electrons, and the Schrödinger model cannot account for the electron-electron interactions. Therefore, it does not give the correct energy for the electrons in a Helium atom. The electron-electron interactions can be viewed as shielding effects. This means that the first electron shields part of the nuclear charge from the second electron. Therefore, the second electron experiences a reduced Coulomb force from the nucleus. Because of the reduced Coulomb force, the binding energy is smaller. It is reduced from -54.4 eV to -24.6 eV. It should be pointed out that the two electrons in the 1s orbital of the He atom are indistinguishable, that means they both have the reduced binding energy. The binding energy only increases to -54.4 eV after one of the two electrons has been removed.

The net positive charge from the nucleus after accounting for shielding effects is called the effective nuclear charge. For the helium atom the effective nuclear charge is 1.34. We can calculate the effective nuclear charge from the experimentally measured first ionization energy. Solving the equation for Z_{eff} gives $Z_{\text{eff}} = 1.34$ (Eq. 1.2.26).

$$E = -\frac{Z_{\text{eff}}^2}{n^2} \times 13.6 \text{ eV} = -24.6 \text{ eV} \quad \text{b/c } IE(\text{He}) = 24.6 \text{ eV}$$

$$Z_{\text{eff}} = \sqrt{n^2 \times \frac{24.6 \text{ eV}}{13.6 \text{ eV}}} = \sqrt{1^2 \times \frac{24.6 \text{ eV}}{13.6 \text{ eV}}} = 1.34$$

Equation 1.2.26 Calculation of Z_{eff} in the Helium atom from first ionization energies.

The Lithium Atom

Now let us go to the atom with three electrons, the lithium atom. Following the so-called Aufbau principle we would fill the first two electrons into the 1s orbital, because the 1s orbital is the orbital with the lowest energy. However, the third electron does not fit into the 1s orbital because of the Pauli principle.



Figure 1.2.38 Wolfgang Pauli (1900-1958), Nobel prize 1945 (Attribution: Nobel foundation [Public domain], commons.wikimedia.org/wiki/File:Pauli.jpg, PD-US-expired)

The Pauli principle states that no two electrons in an atom can have the same four quantum numbers. For that reason an orbital cannot accommodate more than two electrons. Within an orbital two electrons must have different spin quantum numbers s . We can indicate this by writing the electrons as arrows pointing up and down into a square box which represents the orbital. The third electron of the lithium would need to go to the orbital with the next higher energy.

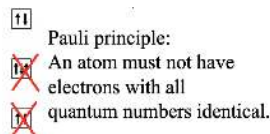
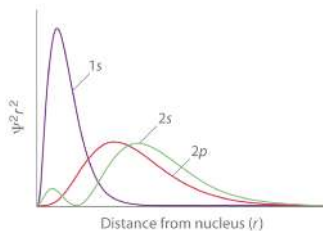


Figure 1.2.39 Illustration of the Pauli Principle

According to the Schrödinger model, the energy of an orbital is only a function of the quantum number n , and we would have two choices: The 2s and the 2p orbitals. Both of them would have the same energy. However, in the lithium atom the 2s orbital has a somewhat lower energy than the 2p orbital. This can again be explained by shielding effects. The 2s orbital penetrates the 1s orbital slightly better than the 2p orbitals. Because of that, the 1s orbital shields the nuclear charge from the nucleus less for the 2s orbital compared to the 2p orbitals. The ground state electron configuration, which is the atom electron configuration having the lowest energy, must therefore be $1s^2 2s^1$. We can see that the 2s orbital penetrates the 1s orbital slightly better than the 2p orbital from the graph for the radial probability functions of the orbitals shown in Figure 1.2.40.



Graph of radial probability functions of the 1s, 2s, and 2p orbitals (Attribution: Inorganic_Chemistry/Map%3A_Inorganic_Chemistry_(Miessler%2C_Fischer%2C_Tarr)/2%3A_Atomic_Structure/2.2%3A_The_Schr%C3%B6dinger_equation%2C_particle_in_a_box%2C_and_atomic_wave_functions/by-nc-sa/4.0/)

The second large maximum of the function of the 2s orbital is actually further away from the nucleus compared to the maximum associated with the 2p orbital. However, because the 2s orbital has a small second maximum very close to the nucleus, the 2s orbital overall penetrates the 1s orbital better, and therefore the shielding effect of the 1s orbital on the 2s orbital is smaller compared to the 2p orbitals.

Generally we can say, the smaller the quantum number l for a given quantum number n , the better the penetration ability of this orbital. Due to the better penetration ability, the shielding is less and thus the effective nuclear charge that acts on the electron in the orbital is higher. The higher effective nuclear charge leads to a lower energy of this orbital. For that reason the energy sequence of orbitals with the same quantum number n increases from s to p to d to f .

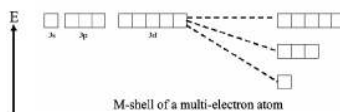


Figure 1.2.41 The energy sequence of orbitals with the quantum number $n=3$ in a multi-electron atom.

For example, a 3s orbital has a lower energy than a 3p orbital which has a lower energy than a 3d orbital (Fig. 1.2.41).

Slater's Rules

Because an orbital energy can be calculated from the effective nuclear charge, it would be useful if the effective nuclear charge could be somehow estimated by simple approximations.



Figure 1.2.42 John C. Slater (1900-1976) (Attribution: Biographical Memoirs, National Academy of Sciences, vol. 53, 1982., <http://scarc.library.oregonstate.edu/coll/pauling/bond/pictures/portrait-slater.html>)

The so-called Slater rules, named after their developer John C. Slater (Fig. 1.2.42), are a simple tool that give a good estimate for the effective nuclear charge of orbitals in multi-electron atoms. The Slater rules estimate a shielding constant σ that allows to calculate the effective nuclear charge Z_{eff} from the nuclear charge according to $Z_{\text{eff}} = Z - \sigma$.

The rules for the estimation of σ are the following. Firstly, we need to recognize that the rules for s and p electrons are slightly different than those for d and f electrons. This is because s and p electrons of the same quantum number n have somewhat higher penetration abilities compared to d and f electrons. For both groups the first step is the same. We write out the electron configuration of the atom according to their quantum numbers n and l as shown below (Fig. 1.2.43).



Figure 1.2.43 First step to calculate the shielding constant σ

Note that this order does not exactly reflect the order of energy. Electrons that have similar shielding effects are grouped together in parentheses. You can see that s and p orbitals of the same quantum number n build a group, all other orbitals are their own group.

To estimate the shielding constant for an s or a p orbital we can now apply the following rules:

- Electrons to the right of the (ns, np) group contribute nothing to σ . This is because these electrons are further away from the nucleus than the electron of consideration.
- Each of the other electrons in the same (ns, np) group contribute 0.35 to σ . These electrons are in the same shell as the electron of consideration, and thus have similar distance to the nucleus. For this reason their shielding is modest only and can be estimated to be about 35% of a full elementary charge, or 0.35.
- Each electron in the $n-1$ shell contributes 0.85 to σ . Those electrons are significantly closer to the nucleus compared to the electron of consideration. For that reason they shield significantly better, approximately 85% of an elementary charge, or 0.85.
- Each electron in the $n-2$ shell or below contributes 1.00 to σ . Those electrons are much closer to the nucleus than the electron of consideration and can fully shield an elementary charge. Its contribution to the shielding constant is therefore approximated to be 1.

Now let us look at the Slater rules for d and f electrons.

- Electrons to the right of the (nd) or (nf) group contribute nothing to σ . This is again because the electron is further away from the nucleus than the electron of consideration, and thus cannot contribute to shielding.
- Each of the other electrons in the same (nd) or (nf) group contribute 0.35 to σ . This is again because these electrons have similar distance compared to the electron for which we calculate the shielding constant, and thus the shielding is modest.
- Each electron to the left of the (nd) or (nf) group contributes 1 to σ . Those electrons are considered much closer to the nucleus, and thus they shield a full elementary charge.

Note that the values 0.35, 0.85, and 1 have been chosen so that the results from the Slater rules are in the best possible accordance with experimental measurements of orbital energies. Remember, when we discussed the helium atom we said that the measurement of ionization energies can provide experimental orbital energies.

Slater's Rules Applied to Multi-Electron Atoms

Let us practice the Slater rules by a few examples. We can, for instance, calculate the effective nuclear charge that acts on a 2p electron in an oxygen atom. To answer this question we need to first write out the electron configuration according to the quantum numbers n and l and group the orbitals correctly (Fig. 1.2.44).

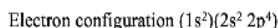


Figure 1.2.44 Slater electron configuration of oxygen

Next, we need to consider that the effective nuclear charge Z_{eff} is the nuclear charge Z minus the shielding constant σ . Then, because oxygen has eight protons, the nuclear charge is 8. From that we need to subtract the shielding constant. Firstly, we need to realize that there are 5 electrons in the same group as the 2p electron of consideration. These are the two 2s electrons and the three other 2p electrons. These five electrons shield with a factor of 0.35 because they are in the same group as the 2p electron for which we want to calculate the shielding constant. Note that this fourth 2p electron does not get a factor because it is the electron for which we calculate the shielding constant. In addition, we need to consider the two 1s electrons. They are in an $n-1$ shell, therefore they contribute with a factor of 0.85. Overall, the effective nuclear charge on the 2p electron is $8 - (2 \times 0.85) - (5 \times 0.35) = 4.55$ (Fig. 1.2.45).

$$\begin{aligned} Z^* &= Z - \sigma \\ &= 8 - [5 \times (0.35)] - [2 \times (0.85)] = 4.55 \\ &\quad (2s, 2p) \quad (1s) \end{aligned}$$

Figure 1.2.45 Effective nuclear charge on the 2p electron of an oxygen atom

We can also ask what is the effective nuclear charge on a 1s electron in the oxygen atom. It is $8 - 0.35 = 7.65$ (Fig. 1.2.46).

$$\begin{aligned} Z_{\text{eff}} &= 8 - 0.35 = 7.65 \\ &\quad (1s) \end{aligned}$$

Figure 1.2.46 Effective nuclear charge on the 1s electron of an oxygen atom

This is because we only need to consider the second electron in the 1s orbital to calculate the shielding constant σ . It contributes with a factor of 0.35 because it is in the same group as the 1s electron of consideration. The 2s and the 2p electrons are in a group right to the 1s orbital, therefore they do not contribute to the shielding.

Slater's Rules and the Aufbau Principle

The Slater rules are of great help to understand the so-called Aufbau principle, which says that for an atom in the ground state the electrons are in the orbitals of the lowest energy. To some extent the orbitals energies follow the quantum number n , but this is not always the case. For example, the element potassium has the electron configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^0 4s^1$ and not $1s^2 2s^2 2p^6 3s^2 3p^6 3d^1$. This means that the energy of the 3d orbital must be higher than the energy of the 4s orbital. Can the Slater rules predict that? Let us therefore calculate the energy of the 3d and 4s orbital using the Slater rules, and see if the energy of the 3d orbital comes out higher than the electron of the 4s orbital.

For the 3d orbital, the shielding constant is $18 \times 1 = 18$ because there are 18 electrons besides the 3d electron of consideration, and all the 18 electrons are in a lower shell. Therefore, they all contribute with a factor of 1. The nuclear charge Z of the potassium is 19. Therefore, the effective nuclear charge $Z_{\text{eff}} = 19 - 18 = 1$ (Fig. 1.2.47).

$$\sigma = 18 \times 1 \rightarrow Z_{\text{eff}} = 19 - 18 = 1$$

Figure 1.2.47 Calculation of effective nuclear charge for the 3d electron of potassium

Now let us do the analogous calculation for the 4s electron. For the 4s electron the Slater rules for s electrons apply. According to that, the two 3s and the six 3p electrons contribute 0.85 to the shielding constant because they are in an $n-1$ shell. The remaining two 1s, two 2s, and six 2p electrons are in an $n-2$ or lower shell and therefore contribute with the factor of 1. Therefore, the overall shielding constant is $8 \times 0.85 + 10 \times 1 = 16.8$. The effective nuclear charge Z_{eff} is therefore $19 - 16.8 = 2.2$ (Fig. 1.2.48).

$$\sigma = 8 \times 0.85 + 10 \times 1 = 16.8 \rightarrow Z_{\text{eff}} = 2.2$$

Figure 1.2.48 Calculation of the effective nuclear charge for the 4s electron of potassium

We can see that a higher effective nuclear charge acts on a 4s electron compared to a 3d electron, which indicates that the orbital energy for the 4s electron will be lower compared to the 3d electron. We can calculate the orbital energies by using the formula $E = -(Z_{\text{eff}}^2/n^2) \times 13.6 \text{ eV}$ (Eq. 1.2.27).

$$E = -\frac{Z_{\text{eff}}^2}{n^2} \times 13.6 \text{ eV}$$

Equation 1.2.27 Equation for orbital energies

For the 3d electron we insert $Z_{\text{eff}}=1$ and $n=3$ which gives -1.51 eV , Eq. 1.2.28.

$$E(3d) = \frac{-1^2 \cdot 13.6 \text{ eV}}{3^2} = -1.51 \text{ eV}$$

Equation 1.2.28 Equation for orbital energy of the 3d electron of potassium

For the 4s electron insert $Z_{\text{eff}}=2.2$ and $n=4$ which gives -4.14 eV .

$$E(4s) = \frac{-2.2^2 \cdot 13.6 \text{ eV}}{4^2} = -4.14 \text{ eV}$$

Equation 1.2.29 Equation for orbital energy of the 4s electron of potassium

As expected, the energy of the 4s electron is lower than the energy of the 3d electron. This explains why the electron configuration with the 4s electron as the valence electron is preferred over that with the 3d electron as the valence electron. Overall, we can see that the Slater rules can correctly predict electron configurations of atoms and the Aufbau principle.

Let us do another, more complex example: There are two conceivable electron configurations for Fe. One in which the 4s subshell is filled before the 3d subshell, and one in which the 3d subshell is filled before the 4s subshell. For the first case the electron configuration is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$. For the second case, the second electron configuration is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^0$. We can see that the two electron configurations are the same for the 1s, 2s, 2p, 3s, and 3p electrons. We therefore focus on the remaining eight electrons, denoted in red, and calculate their energies.

In this case, we need to compare the sum of the electron energies for both electron configurations to be able to decide which electron configuration is favored. We calculate the sum of the energies of the eight electrons for the first electron configuration the following way: In the first step we need to rewrite the electron configuration in Slater form: $(1s^2)(2s^2 2p^6)(3s^2 3p^6)(3d^6)(4s^2)$.

Next, we can calculate the shielding constant for a 3d electron. We can ignore the two 4s electrons because they are to the right of the d electron of consideration, and do not contribute to the shielding. However, we do need to consider the five other 3d electrons that are in the same group as the 3d electron of consideration. They contribute with the factor of 0.35. The other 18 electrons contribute 1.00 because they are in a lower shell. Because the effective nuclear charge of Fe is 26, the effective nuclear charge $Z_{\text{eff}} = 26 - 19.75 = 6.25$ (Eq. 1.2.30).

$$\sigma_{3d} = 5(0.35) + 18(1.00) = 19.75 \quad Z_{\text{eff}}(3d) = 26 - 19.75 = 6.25$$

Equation 1.2.30 Equation for the shielding constant and effective nuclear charge of the 3d electron of an iron atom

For a 4s electron, there is one other 4s electron in the same group which contributes with a factor of 0.35. In addition, there are 14 electrons in an $n-1$ shell, namely the 3s, the 3p, and the 3d electrons. They contribute 0.85 to the shielding constant. The remaining 10 electrons are in an $n-2$ shell or lower, therefore they contribute with the factor 1.00. This gives an overall shielding constant of 22.25. From that we can calculate the effective nuclear charge for a 4s electron which is $26 - 22.25 = 3.75$.

$$\sigma_{4s} = 1(0.35) + 14(0.85) + 10(1.00) = 22.25 \quad Z_{\text{eff}}(4s) = 26 - 22.25 = 3.75$$

Equation 1.2.31 Equation for shielding constant and effective nuclear charge of the 4s electron of an Fe atom

We can now calculate the electron energies. It is the sum of the energies for the 3d and 4s electrons. The electron energy for a single electron is $E = -(Z_{\text{eff}}^2/n^2) \times 13.6 \text{ eV} = -(6.25^2/3^2) \times 13.6 \text{ eV} = -59.03 \text{ eV}$. Because we have six 3d electrons, the overall energy for the six electrons is $6 \times -59.03 \text{ eV} = -354.17 \text{ eV}$ (Eq. 1.2.32).

$$E(3d) = -\frac{Z_{\text{eff}}^2}{n^2} \times 13.6 \text{ eV} = -\frac{6.25^2}{3^2} \times 13.6 \text{ eV} = -59.03 \text{ eV}$$

$$\text{For six 3d electrons: } E(3d) = 6 \times -59.03 \text{ eV} = -354.17 \text{ eV}$$

Equation 1.2.32 Equation for the electron energies of the six 3d electrons in an iron atom

The energy of a single 4s electron is $E = -(3.75^2/4^2) \times 13.6 \text{ eV} = -11.95 \text{ eV}$. For two 4s electrons the energy is $2 \times -11.95 \text{ eV} = -23.90 \text{ eV}$.

$$E(4s) = -\frac{3.75^2}{4^2} \times 13.6 \text{ eV} = -11.95 \text{ eV}$$

$$\text{For two 4s electrons: } E(4s) = 2 \times -11.95 \text{ eV} = -23.90 \text{ eV}$$

Equation 1.2.33 Equation for the electron energies of the 4s electron in iron

The sum of all electrons is then $-354.17 \text{ eV} + (-23.90 \text{ eV}) = -378.1 \text{ eV}$ (Eq. 1.2.34).

$$E(4s + 3d) = -354.17 \text{ eV} - 23.90 \text{ eV} = -378.1 \text{ eV}$$

Equation 1.2.34 Equation for the sum of all electrons

Now let us calculate the energy for the second electron configuration. First we need this electron configuration in Slater configuration. Next, we can calculate the shielding constant for the d electrons. It is $\sigma_{3d} = 7(0.35) + 18(1.00) = 20.45$ because there are seven other 3d electrons in the same group and 18 electrons in lower shells. The effective nuclear charge is then $Z_{\text{eff}}(3d) = 26 - 20.45 = 5.55$.

$$\sigma_{3d} = 7(0.35) + 18(1.00) = 20.45$$

$$Z_{\text{eff}}(3d) = 26 - 20.45 = 5.55$$

Equation 1.2.35 Equation for the shielding constant and effective nuclear charge of the 3d electrons of iron (second configuration)

The sum of the electron energies for the eight d electrons is then $E(\Sigma 3d) = -8(5.55^2/3^2) \text{ eV} = -372.3 \text{ eV}$, Eq. 1.2.36.

$$E(\Sigma 3d) = -8 \frac{5.55^2}{3^2} \text{ eV} = -372.3 \text{ eV}$$

Equation 1.2.36 Equation for the sum of the electron energies for the eight d electrons of iron

We can see the energy of the second electron configuration is somewhat higher, and thus less preferred. This is in accordance with the experimentally observed ground state electron configuration of the Fe atom. Again, we see that the Slater rules can correctly predict the ground state electron configuration of atoms.

The Aufbau Principle and the Spin Pairing Energy

The Slater rules are suitable to account for shielding effects in multi-electron atoms which are of electrostatic nature. However, not only electrostatic effects influence the orbital energies, there are also magnetic effects. This is because each electron has a spin with a magnetic field associated with it. The electrons behave like little magnets that can interact with each other. These interactions can be either attractive or repulsive, which influences the electron energies. The electron energies are usually minimized when the number of electrons with the same spin is maximized. This is known as Hund's (Fig. 1.2.49) rule of maximum spin multiplicity.



Figure 1.2.49 Friedrich Hund (1898-1997) (Attribution: GFHund [CC BY (<https://creativecommons.org/licenses/by/3.0/>)], commons.wikimedia.org/wiki/File:_%B6ttingen.jpg)

In the previously example we have seen that the electron configuration $4s^2 3d^6$ is favored in Fe because of electrostatic shielding effects. In addition, it is also favored by magnetic effects. In this electron configuration there are four unpaired electrons, while in the electron configuration $3d^8$ there are only two unpaired electrons.



Figure 1.2.50 Electron configuration of Fe depicted regarding spin pairing

We can now understand the Aufbau principle for multi-electron atoms (Figure 1.2.51).

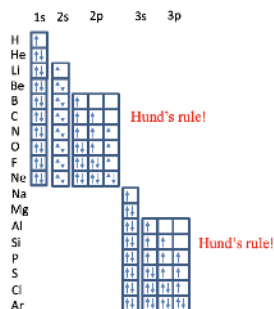


Figure 1.2.51 Electron configuration of atoms following the Aufbau principle

For H and He the 1s orbital gets filled. From lithium to beryllium the 2s subshell gets filled, and from boron to neon the 2p subshell gets filled under consideration of Hund's rule. You can see that for carbon the two 2p electrons are both spin up in different 2p orbitals. For nitrogen there are three unpaired electrons in the 2p orbitals. From the element oxygen on, we have to start pairing spins until all spins are paired in neon. From Na to Ar the 3s and the 3p orbitals get filled according to the same principles.

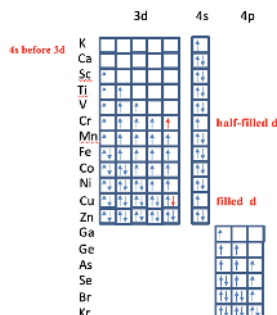


Figure 1.2.52 Continuation of electron configuration of atoms following the Aufbau principle

After that, the 4s orbitals of K and Ca get filled, and then the 3d orbitals of the elements from Sc to Zn (Fig. 1.2.52). We have seen previously that the Slater rules nicely show why. You can see that there are two anomalies though. For the chromium, the electron configuration is $4s^1 3d^5$ and not $4s^2 3d^4$. This is because there are more unpaired spins in this electron configuration, and a half-filled 3d subshell with unpaired spins only represents a particularly stable electron configuration. The second anomaly occurs for the element Cu which has an electron configuration $4s^1 3d^{10}$ and not $4s^2 3d^9$. This is because a d^{10} electron configuration is a particular stable electron configuration. After the 3d subshell is filled, the 4p subshell is filled for the elements from Ga to Kr. Again, the electrons are filled into the orbitals according to Hund's rule. We could extend our consideration to even heavier elements, but will be beyond scope.

Dr. Kai Landskron ([Lehigh University](#)). If you like this textbook, please consider to make a donation to support the author's research at Lehigh University: [Click Here to Donate](#).

This page titled [1.2: The quantum-mechanical model of the atom](#) is shared under a [not declared](#) license and was authored, remixed, and/or curated by [Kai Landskron](#).

- [1.2: The quantum-mechanical model of the atom](#) by Kai Landskron is licensed [CC BY 4.0](#).

1.3: Periodic Properties of Atoms

The Periodic Table of Elements

Having understood the Aufbau principle we can now understand the periodic table of the elements. The periodic table was actually invented a long time before the quantum-mechanical atom model was developed. It was first introduced by Mendeleev and Meyer who ordered the elements according to their masses in a row. Whenever an element had a property similar to an element that already was previously considered a new row was begun. This gave a table with rows and columns. Within a row the mass of the elements would increase, within a column one would find the elements that had similar properties. The rows would be called periods, and the columns would be called groups, hence the name periodic table of the elements. At the time Mendeleev and Meyer invented the periodic table, there were still many elements undiscovered visible as “holes” within the periodic table. However, over time the periodic table became complete (Fig. 1.3.1).

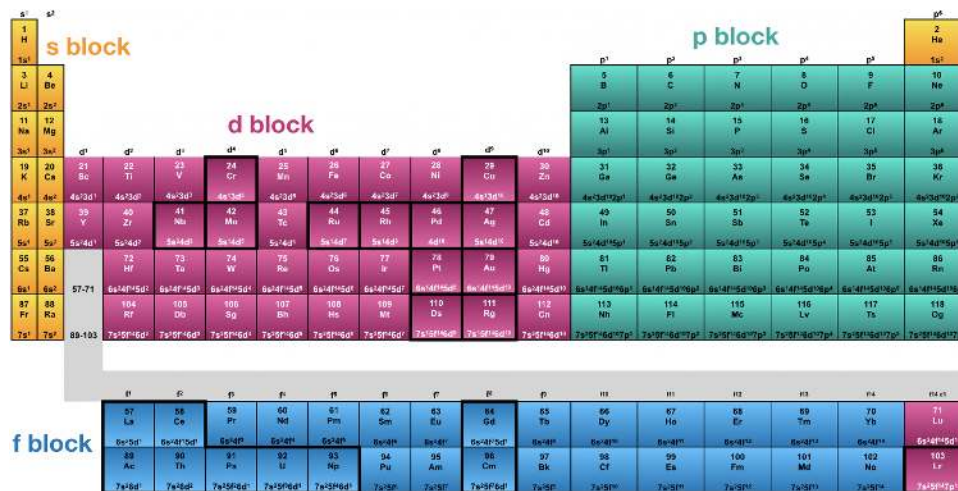


Figure 1.3.1 The periodic table of elements (Attribution: Kathryn Haas https://chem.libretexts.org/Bookshel...bau_Principle#)

Today we know 118 elements completing seven periods and 18 groups. Today, the elements are not ordered by mass, but by the atomic number (number of protons), however this did not change the original periodic table much because except for a very few elements the mass of the elements follows the atomic number. Before the quantum-mechanical model was developed, it was not understood why elements within the same group had similar properties. When you analyze the electron configuration of the atoms you can see that all atoms in the same group have the same type of outmost electrons, also called valence electrons. These electrons are the electrons that have the greatest distance from the nucleus and have the highest energies. Because they have the highest energies, they are also the electrons that are most reactive, and determine the chemical and physical properties of the element. For example, in the first group all elements have one s electron as the valence electron. Because these electrons have the highest energy and are most reactive, they determine the properties of the atom, and thus the elements in the first group have similar properties.

When you look into group 2, all the electrons have two s electrons as the valence electrons. An exception is the element helium which is placed into group 18 which is called the group of the noble gases. The other group 18 elements have two s and six p electrons of the same quantum number n as valence electrons. The behavior of He is much more similar to that of the other group 18 elements, namely it is a gas with extremely low reactivity. This property is due to the fact that like the other noble gases He has a full shell. Elements with a full shell are particularly unreactive because full shells represent particularly stable electron configurations, also called noble gas configurations.

Group 1, group 2 and He are called the s-block of the periodic table because they are the elements that only have s electrons as valence electrons. From group 3 to group 12 there are elements with s and d electrons as valence electrons. These elements are the d-block of the periodic table. The d-block begins with group 3 where there is only one d electron together with two s electrons, and ends with group 12 where there are ten d and two s electrons. In group 12 the d subshell is completely filled with electrons. From group 13 to group 18 we find elements that have p electrons as valence electrons. There are six groups because p orbitals can accommodate up to six p electrons. In group 13 there is only one p valence electron, in group 14 there are two, in group 15 there are three, in group 16 four, in group 17 five, and in group 18 there a six except the Helium which we previously discussed already. Group 13 to 18 is called the p block of the periodic table. The s-block and the p block elements together are also called the main

group elements, the elements of the d-block are called the transition metals. The elements of the main groups have specific names. Group 1 elements except hydrogen are called the alkali metals, group 2 elements the earth alkaline metals, group 13 elements the triels, group 14 elements the tetrels, group 15 elements the pnictides, group 16 elements are the chalcogens, group 17 elements the halogens, and group 18 elements the noble gases.

Finally, there is the f block of the periodic table. The f-block contains the elements with f-valence electrons. Because there are seven f-orbitals for a given quantum number n , the f orbitals can accommodate up to 14 electrons, and thus there are 14 groups. The f block is actually located in between the s block and the d block of the periodic table, but is typically written underneath the rest of the periodic table. This is done because the periodic table would become too wide if it was placed between the s and the d block. Note that the f-block elements do not have a group number. Note also that the elements on the left end of the f-block, lanthanum and actinium, actually do not have f-valence electrons. They are actually group 3 elements located underneath the elements scandium and yttrium. The first row of the f-block is are called the lanthanides, which is the greek meaning the elements following the lanthanum. Only the elements following the lanthanum have f-valence electrons. Analogously, the actinium is not an actual f-block element, only the elements that follow the actinium. They are called the actinides.

The quantum mechanical model of the atoms also explains why the periodic table has periods. Each period is associated with a quantum number n , the first period with $n=1$, second period with $n=2$ and so forth. Within a period the available s and p valence orbitals of a specific quantum number n are getting filled. Subshells of a lower quantum number may also be filled within the period. For example, the 3d subshell gets filled within the fourth period, and the 4f subshell gets filled in the 6th period.

Elements with electron configurations that deviate from the expected electron configuration are highlighted in bold (Figure 1.3.1). There are only exceptions for the d-block and f-block elements. We discussed some of them already, for example the Cr and the Cu. In addition, for instance, Pd has the electron configuration $4d^{10}$ instead of $5s^2 4d^8$, and Pt has the electron configuration $6s^1 5d^9$ instead of $6s^2 5d^8$. We can explain these exceptions when considering the complex electrostatic and magnetic interactions of electrons in multi-electron atoms. Not everything can be explained by the simple concepts of Slater rules and Hund's rule.

As chemists we should always know our periodic table. Learning is easier with music, and there is a fabulous song by Tom Lehrer that can help you to learn the periodic table. The link to a respective YouTube video is provided below.

<https://www.youtube.com/watch?v=DYW50F42ss8>

Atomic Radii

Let us now look at what is called the periodic properties of atoms. These are properties which change due to the position of the element within the periodic table. Let us first consider how the atomic radii of the atoms depend on the position of the atom in the periodic table, Fig. 1.3.2.

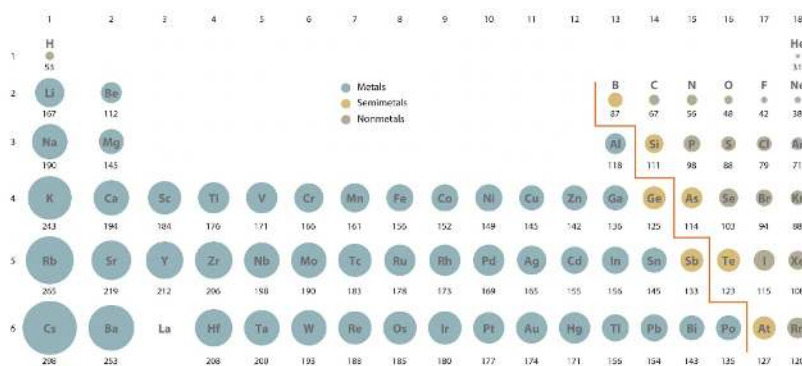


Figure 1.3.2 Calculated atomic radii of the elements in the periodic table (Attribution: Webelements.com, <https://chem.libretexts.org/@api/deki/files/27875/bd05f43d0392ab934fc21044ccca1cfd.jpg?revision=1> <http://creativecommons.org/licenses/by-nc-sa/3.0/us/>)

When you look at how the atomic radii change within a period you can easily see that they decrease from the left to the right in the periodic table. The decrease is more significant for the s and p block elements compared to the d block elements. We can explain this effect using the concept of effective nuclear charge. As we go from the left to the right in the periodic table we add electrons to the same shell. Because also a proton is added to the nucleus with each electron added, the electron experiences a greater attractive force from the nucleus. This effect tends to increase the effective nuclear charge. However, there is also an opposite effect. Because there is one more electron, there are more electron-electron repulsion and shielding effects, and this tends to decrease the effective

nuclear charge. Because the new electron is added to the same shell, the shielding effects do not fully compensate for the additional Coulomb force coming from the additional proton in the nucleus, and thus the effective nuclear charge increases. This higher effective nuclear charge pulls the electrons closer to the nucleus, and thus the atom size decreases. For the d-block elements new d electrons get added to a more inner subshell, for example for the d-block elements of the 4th period the d electrons get added to the 3d subshell. Because of that, these 3d electrons can more effectively shield nuclear charge from the outmost two 4s electrons, and thus the decrease of the atomic radius is less pronounced. In summary:

Trend 1: Atomic radius decreases continuously within a period.

A second trend is that each time a new period is begun the atomic radius jumps dramatically. For example it jumps from 31 pm to 167 pm as we go from the helium to the lithium, and from 38 pm to 190 pm as you go from the neon to the sodium. This trend is because the electrons get filled into a new, more outer shell when a new period is begun. This outer shell is located much farther away from the nucleus, and thus the radius of the atom increases. In addition, the electrons in the inner shells can very effectively shield the nuclear charge from the electron added to the new shell which tends to increase atom radius. In summary:

Trend 2: Atomic radius jumps when a new period is begun.

As a third trend you can see that the atomic radii increase as you go down a group. This is because within a group there are the same number and type of valence electrons, but the valence electrons are in a more outer shell associated with a higher quantum number n . In a somewhat different view we can also argue that the first two trends add up to give the third trend. The effect of jump of atomic radius when a new period is begun is greater than the effect of decline of the radius within a period. As a result, the radius tends to increase as you go down a group. In summary:

Trend 3: Atomic radius increases continuously within a group.

The determination of an atomic radius is not as trivial as it may seem. You can obtain it either from quantum mechanical calculations, or determine it experimentally. If you calculate it, then the radius is defined as the radius that defines the volume in which the electron can be observed with 90% probability. The radii that you see in Figure 1.3.2 are actually calculated radii. If you determine the radius experimentally, then there are again different subtypes of radii. The so-called metallic, or crystal radii, the van der Waals radii, and the covalent radii. The covalent radius is the half of the distance between two, same atoms in a molecule held together in a single covalent bond.

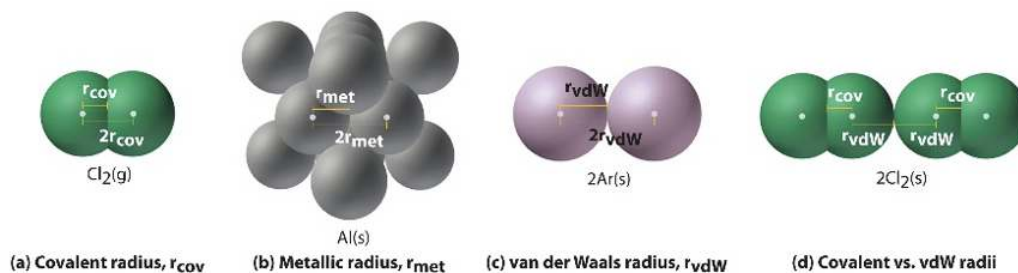


Figure 1.3.3 Atomic radii for a) covalent radius b) metallic radius c) van der Waals radius d) covalent vs vdW radii (Attribution: Chemlibretexts. <https://chem.libretexts.org/@api/dek...jpg?revision=1> <http://creativecommons.org/licenses/by-nc-sa/3.0/us/>)

This is unambiguous for molecules such as Cl_2 , the other halogens, and for other cases such as hydrogen, carbon (as diamond), sulfur, and a few other cases. However for oxygen, O_2 , the situation is less clear as the order of the oxygen-oxygen bond is 2. In this case, it is necessary to infer the covalent radius from molecules containing O-O single bonds or from molecules containing an E-X single bond (E=element) in which the covalent radius of X is known. The van der Waals radius is half the distance between two same atoms in a crystal when there is no actual bonding between the atom except weak van der Waals forces. For example, in solid argon, there are only van der Waals forces between the atoms, and half the distance between the atoms in solid argon is the Van der Waals radius. For chlorine, for example, the van der Waals radius is half the distance between two chlorine atoms of two different Cl_2 molecules in solid Cl_2 . Note that the van der Waals radius of chlorine is different than the covalent radius of chlorine. This again shows that there is a certain ambiguity when determining an atomic radius, and therefore it is always important to say what definition of atomic radius you use. Lastly, there is also the metallic radius that applies to solid metals in which metal atoms are held together by metallic bonding. The metallic radius is defined as half the distance between the centers of two metal atoms within a metal.

First Ionization Energy

The first ionization energy is another important periodic property of the elements. It is defined as the energy required to remove an electron from a neutral atom in the gas phase. It is important to understand that we refer to an atom in the gas phase here, because only in the gas phase atoms do not significantly interact with each other. Therefore, only when looking at an atom in the gas phase we can truly determine the properties of single, isolated atoms. Let us look at the two images (Figure 1.3.4) to determine periodic trends.

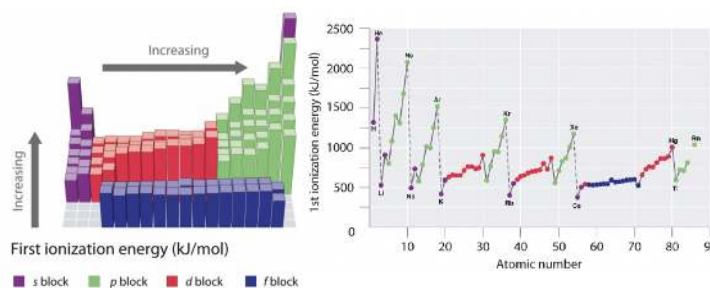


Figure 1.3.4 First ionization energies as a third dimension in the periodic table (left) and as function of atomic number (right)
(Attribution: ChemlibreTexts <https://chem.libretexts.org/@api/dek...png?revision=1> <http://creativecommons.org/licenses/by-nc-sa/3.0/us/>)

First of all we need to realize that all 1st ionization energies are positive. This means that it always requires energy to remove an electron from a neutral atom. You can understand this when you consider that all electrons in an atom are bound to the nucleus via a binding energy. If an ionization energy was negative then this would mean that the atom would spontaneously lose an electron, and spontaneously ionize. This would be unreasonable.

Now let us see how the first ionization energy changes within a period. We can see that it increases as we go from the left to the right within a period. This trend can be explained by the fact that the effective nuclear charge increases from the left to the right within a period. As the effective nuclear charge increases, the pull of the nucleus on the outmost electron increases, and thus the harder it is to remove this electron from the atom. You can see, however, that within the main group elements the trend is two times broken. The group 13 elements have a lower ionization energy than the group 2 elements, and the group 16 elements have a lower ionization energy than the group 15 elements. This phenomenon can be explained by the fact that the filled and half-filled subshells in group 2 and group 15 represent particularly stable electron configurations, and thus the electrons within these subshells have unusually low energy. For that reason these electrons are harder to remove from the atoms. The group 2 elements have filled s subshells, and the group 15 elements have half-filled p subshells. You can also notice that the increase in ionization energy is less pronounced within the d-block, and even less pronounced within the f-block. This is because the effective nuclear charge on the outmost electrons does not increase as much because inner d and f orbitals are getting filled. You can also see that the group 13 elements have a lower ionization energy than the group 12 elements. This is because in group 13 a new p subshell is begun and the new electrons get added to the outmost shell where shielding effects are smaller.

As a second trend you can see that whenever a new period is begun, the ionization energy drastically drops. This is because the new electron is added to a new, more outer shell where the effective nuclear charge acting on the new electron is much smaller. The third trend is that the ionization energy becomes smaller as we go down a group. This is because the effective nuclear charge on the valence electrons decreases because the quantum number n of the valence electrons increases. There is one peculiarity that needs additional explanation. The drop of ionization energy from hydrogen to lithium in the first group is unusually large. This is because hydrogen is the only element for which there are no shielding effects, simply because the hydrogen has only one electron. For that reason the valence 1s electron of the hydrogen is far harder to remove than the valence s electrons of the other group 1 elements.

Electron Affinity

Now let us consider the first electron affinity (Figure 1.3.5). It is defined as the energy required or released when you add an electron to a neutral atom in the gas phase. Again, we look at an atom in the gas phase, because we want to consider an isolated atom that does not make significant interactions with other atoms.

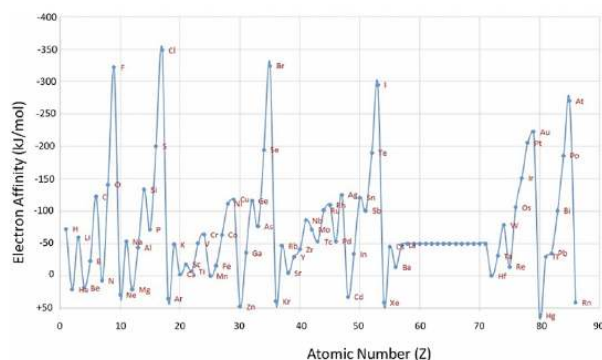


Figure 1.3.5 Electron affinity vs Atomic Number (Attribution: Chemlibretexts. <https://chem.libretexts.org/@api/dek...780&height=456> <http://creativecommons.org/licenses/by-nc-sa/3.0/us/>)

We can see from the graph that most elements have negative electron affinities, meaning that the addition of a free electron to a neutral atom is exothermic, and releases energy, however, this is not always the case. Some elements have negative electron affinities, in particular the noble gases, but also Be, N, Mg, Zn, Cd, and Hg. Zn, Cd, and Hg, are group 12 elements with a full d subshell which is particularly stable. An additional electron would need to be added to a p orbital of a higher shell which is energetically quite unfavorable. Similarly, Be and Mg have filled s subshells which are also fairly stable. Adding an additional electron would need to start a new subshell, which is energetically not favorable. You can see that Ca and Sr have electron affinities of about 0, and that of Ba is only slightly negative showing that higher periods make the addition of an electron to a new p shell slightly more favorable. Nitrogen has a half-filled p subshell, which is also a quite stable electron configuration, and therefore adding an electron is not favorable. The addition of an electron is slightly more favorable for the other group 15 elements, however, these also tend to have rather low electron affinities.

Generally, for the main group elements the electron affinity tends to increase from the group 1 to group 17. However, the trend is broken for group 2 and group 15 elements because these elements have full and half-filled subshells, respectively. In addition, it is noteworthy that group 1 elements have higher electron affinities than group 3 elements. This is because adding an electron to a group 1 element produces a full s subshell which is fairly stable. For the d-block elements, the electron affinity tends to increase from group 3 to group 11, but the trend is broken multiple times showing that each element would need to be investigated individually which is beyond the scope here. There is a big drop in electron affinity from group 17 to group 18 which is easily explained by the fact that the addition of an electron to a group 17 element produces a filled shell, while a group 18 element already has a full shell, and the addition of an electron would start a new shell. Similarly, there is a sharp drop in electron affinity from group 11 to group 12, as an addition of an electron to a group 11 atom produces a full subshell, while the addition of an electron to a group 12 element would require the start of new, more outer subshell. For the p block elements the electron affinity tends to first increase from period 2 to 3, and then decrease. For the s-block there is a steady decrease down a group. For d block elements there is no clear trend for the electron affinity.

Electronegativity

Lastly, let us study the electronegativity as a periodic property. The electronegativity is the measure of the ability of an atom to attract electrons within a chemical bond. It is related to the electron affinity in the sense that both period properties measure the force by which an electron is attracted to a neutral atom. However, the electron affinity is associated with isolated atoms, that do not make bonds to neighbored atoms, while the electronegativity refers to atoms that are bonded to neighbored atoms. The importance of the electronegativity stems largely from its ability to predict and understand the nature of the bonding between atoms, in particular, covalent, ionic, and metallic bonding.

Electronegativity Scales

Electronegativity was a concept first developed by Linus Pauling to describe the relative polarity of bonds and molecules. He argued that the electronegativity of an atom could be derived from bond energy differences between homoleptic and heteroleptic bonds (Eq. 1.3.1).

$$\chi(A) - \chi(B) = c [(E(A-B) - \{E(A-A) \cdot E(B-B)\}^{1/2}), \text{ eV}].$$

Here $E(A-B)$, $E(A-A)$ and $E(B-B)$ are energies of A-B, A-A and B-B bonds respectively.
c is a constant.

Equation 1.3.1 Electronegativity according to Pauling, eV = electron volts.

For example, the energy required to break the bond in H_2 is 432 kJ mol^{-1} , and for F_2 it is 159 kJ mol^{-1} . However the energy required to break a bond in HF is 565 kJ mol^{-1} , which is much higher than expected just by averaging the energy of the two homonuclear bonds (296 kJ mol^{-1}). Pauling argued that the difference could be assigned to electrostatic attractions between the F and H “ends” of the molecule whereby the F end has more electron density and the H end has less electron density. The greater the difference between the average energy of the homoleptic bonds and the heteroleptic bonds, the greater the electronegativity difference between the atoms, and the greater the polarity of the heteroleptic bond.

First, Pauling used the arithmetic means of the heteroleptic bond energies, later he used the geometric means, because he found empirically that it worked better (Equation 1.3.1). The geometric mean of the homoleptic bond energies is the square root of the product of the homoleptic bond energies. This method only provided electronegativity differences. He therefore needed to define a reference atom with an arbitrarily defined electronegativity value, and then determine the electronegativity values of all other atoms relative to that value. He chose the most electronegative atom, the fluorine atom, as the standard and assigned a value of 4.0. The values of all other atoms vary between 0.7 (Fr) and 3.44 (O), and are shown in the periodic table on the below (Fig. 1.3.6).

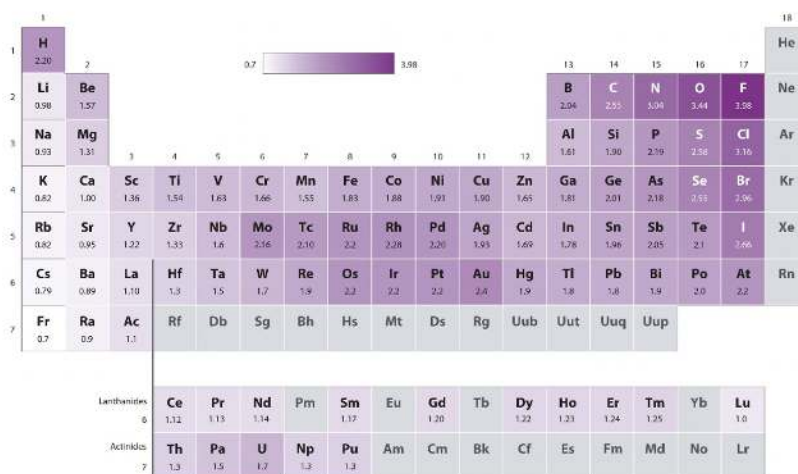


Figure 1.3.6 Periodic trends of electronegativity (Attribution: Chemlibretexts <https://chem.libretexts.org/@api/dek...jpg?revision=1> <http://creativecommons.org/licenses/by-nc-sa/3.0/us/>)

What are the periodic trends? One can see that within a period the electronegativity values of the main group elements strictly increase with the group number. Note though that the noble gases do not have a Pauling electronegativity because, with few exceptions, noble gases do not make compounds and thus no bond energies are available for them. We can also see that for main group elements the electronegativity strictly decreases down a group. This makes the fluorine the most electronegative atom, and the cesium the least electronegative atom. Cesium has an electronegativity value of 0.7. We ignore the radioactive francium here. For the d-block, the trends are less strict, but there is a tendency of electronegativity increase from group 3 to group 11. All group 12 elements have smaller electronegativities compared to their neighbored group 11 elements. What about trends within a group? In group 3 to group 5 there is a small decrease in electronegativity down a group. For group 6 and 9, the period 5 elements have higher values than their neighbored elements in period 4 and 6. For group 10 to 12 the electronegativity increases down a group. The d-block element with the highest electronegativity is the gold, it has a value of 2.4. Note that this is similar to the electronegativity of non-metals such as I, S, and P. For these reasons, sometimes gold can behave like a non-metal in compounds with very low electronegativity. For example, there is the compound cesium auride (CsAu) which is a transparent, ionic crystalline compound due to the high electronegativity difference between cesium and gold. The f-block elements generally have low electronegativity.

The Pauling electronegativity scale is derived from empirical data on bond energies. It works very well in practice and it is up to date the most used electronegativity scale. However, it does not relate electronegativity to other periodic properties, and the quantum-mechanical model of the atoms. One electronegativity scale that addresses this short-coming the Allred-Rochow scale. It relates the electronegativity to the Coulomb force that acts on an electron on the surface of an atom. The Coulomb force is proportional to the effective nuclear charge Z^* and inverse proportional to the atomic radius square. Z^*/r^2 is multiplied with a factor of 3590 and a value of 0.744 is added to this term. These numbers are empirically chosen so that the values of the Allred-Rochow scale become comparable to the Pauling scale.

$$\text{Allred-Rochow: } \chi = [0.744 + 3590 \cdot Z^* / r^2]$$

Here Z^* is effective nuclear charge and r is atomic radius.

Equation 1.3.2 Allred-Rochow definition of electronegativity

Another frequently used scale is the Mulliken electronegativity scale. The Mulliken scale relates the electronegativity to the sum of the first ionization energy IE and the first electron affinity EA. We would intuitively agree that the ability of an atom to attract electrons within a chemical bond is the higher, the harder it is to remove an electron from an isolated atom, and the easier it is to add an electron to an isolated atom. The numbers 0.118 and -0.207 are empirically chosen to make comparisons to the Pauling scale possible. The Mulliken scale is related to the Allred-Rochow scale in the sense that you can explain ionization energies and electron affinities of atoms with the concept of the effective nuclear charge.

$$\text{Mulliken: } \chi = [0.118 \cdot (\text{IE} + \text{EA}) - 0.207]$$

Here IE is ionization energy and EA is electron affinity of an atom

Equation 1.3.3 Mulliken definition of electronegativity

Another electronegativity scale has been developed by Leland C. Allen. It designed for main group elements in particular. It argues that the electronegativity is proportional to the average energy of the s and p valence electrons in the atom. This electronegativity concept also has a relationship to that of Allred-Rochow, because orbital energies can be calculated from the effective nuclear charge.

$$\text{Allen: } \chi = [(m e_p + n e_s) / (m + n)]$$

Here e_p is the energy of p-electrons and e_s is the energy of s-electrons; m and n are populations of p- and s-orbitals respectively.

Equation 1.3.4 Leland C. Allen definition of electronegativity

Overall, you can see that the electronegativity scales are inter-related, they say essentially the same, but present electronegativity from a somewhat different perspective.

Electronegativity of the Elements and Chemical Bonding

One of the most powerful attributes of electronegativity is that it can predict what type of chemical bond is to expect in elements and compounds. Let us first look at the bonding between elements.

PAULING ELECTRONEGATIVITY VALUES																						
1 H 2.20																	13 B 2.04	14 C 2.55	15 N 3.04	16 O 3.44	17 F 3.98	
3 Li 0.98	4 Be 1.57																	19 Al 1.61	20 Si 1.90	21 P 2.19	22 S 2.58	23 Cl 3.16
5 Na 0.93	6 Mg 1.31																	31 Ga 1.81	32 Ge 2.01	33 As 2.18	34 Se 2.55	35 Br 2.96
7 K 0.82	8 Ca 1.00	9 Sc 1.36	10 Ti 1.54	11 V 1.63	12 Cr 1.66	13 Mn 1.55	14 Fe 1.83	15 Co 1.88	16 Ni 1.91	17 Cu 1.90	18 Zn 1.65	29 Ga 1.81	30 Ge 2.01	31 As 2.18	32 Se 2.55	33 Br 2.96						
19 Rb 0.82	20 Sr 0.95	21 Y 1.22	22 Zr 1.33	23 Nb 1.6	24 Mo 2.16	25 Tc 1.9	26 Ru 2.2	27 Rh 2.28	28 Pd 2.20	29 Ag 1.93	30 Cd 1.69	31 In 1.78	32 Sn 1.96	33 Sb 2.05	34 Te 2.1	35 I 2.66						
37 Cs 0.79	38 Ba 0.89	39 La 1.1	40 Hf 1.3	41 Ta 1.5	42 W 2.36	43 Re 1.9	44 Os 2.2	45 Ir 2.20	46 Pt 2.28	47 Au 2.54	48 Hg 2.00	49 Tl 1.62	50 Pb 2.33	51 Bi 2.02	52 Po 2.0	53 At 2.2						
55 Fr 0.7	56 Ra 0.9															81 Tl 1.62	82 Pb 2.33	83 Bi 2.02	84 Po 2.0	85 At 2.2		

Figure 1.3.7 Pauling electronegativity values for the elements (Attribution: Ck12.org https://www.ck12.org/flx/show/default/image/user%3Ack12editor/201411241416847232690011_ab8840393ea93a2a7c9fc01e395da3b2-201411241416847569430107.png <https://creativecommons.org/licenses/by-nc/3.0/legalcode>)

In the depicted periodic table (Figure 1.3.7) you can see the type of chemical bonding between atoms indicated by different colors. There is metallic bonding in metals, indicated by the color blue. In metallic bonds the electrons are shared between atoms but are delocalized over many atoms in the metal. The green color indicates covalent bonding seen in non-metals. In covalent bonding the electrons are shared, but localized in between typically two atoms. Metalloids have chemical bonding with a mix of covalent and metallic character shown in orange. There is electron-sharing with a moderate degree of delocalization. The Pauling electronegativity values of the elements are written underneath the element symbols.

Can we relate the electronegativity values to the chemical bonding in the elements? We can clearly see that elements with high electronegativity values significantly above 2.0, the non-metals, tend to make covalent bonds in between them. All metalloids with a hybrid covalent-metallic character, have intermediate electronegativity around 2.0. Most metals have low electronegativity values

below 2.0 except noble metals like Pt and Au. Overall, we can say that there is a clear relationship between the bonding type in an element and its electronegativity. Now let us see if electronegativity can also predict chemical bonding in compounds.

Ketelaar's Triangle

The ability of electronegativity to predict the bonding type in compounds can be understood by Ketelaar's triangle, named after J.A.A. Ketelaar.

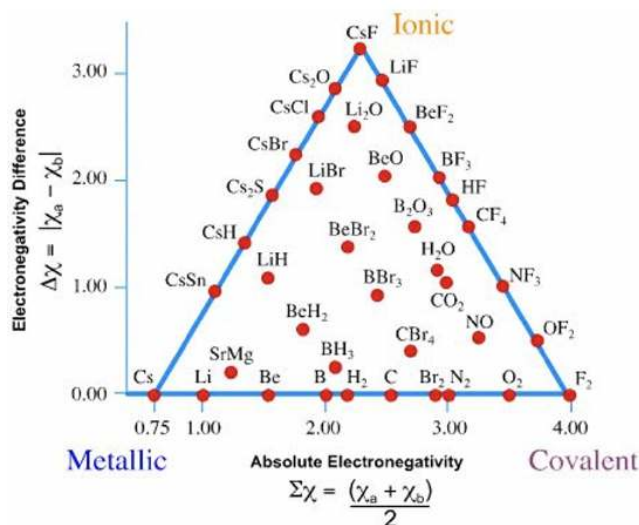


Figure 1.3.8 Ketelaar's Triangle (Attribution: Chemlibretexts <https://chem.libretexts.org/@api/dek...jpg?revision=2>
<http://creativecommons.org/licenses/by-nc-sa/3.0/us/>)

You can see that Ketelaar's triangle has three corners (Figure 1.3.8). Two of the corners are occupied by the elements F and Cs, and the third corner is occupied by CsF. All elements are located on the horizontal edge of the triangle, and there many compound indicated by dots either on the two other edges or within the triangle. There are no compounds or elements outside of the triangle. How can we understand this triangle and what does it say about the relationship between bonding character and electronegativity? To understand this, look at the two axes. The horizontal x-axis represents the average electronegativity of the atoms in an element or a compound. The vertical y-axis represents the electronegativity difference between the atoms in the elements or the compound. For elements the electronegativity difference between the atoms is zero, because in an element all atoms are of the same type. Therefore, all the elements lie on the horizontally oriented edge of the triangle. Because Cs is the element with the lowest electronegativity, it lies furthest to the left on this edge. Fluorine is the element with the highest electronegativity, and thus lies the furthest on the right side. The other elements are in located in between the cesium and the fluorine on the edge. The higher the electronegativity of the elements, the further right their position on this edge of the triangle.

For compounds, the electronegativity difference between atoms is never exactly zero, therefore all compounds are located above the horizontal edge of the triangle. The compound with the highest electronegativity difference is the CsF. Its average electronegativity is the sum the electronegativity of Cs and F divided by 2: Therefore, CsF defines the third corner of the triangle. All other compounds must lie on the edges or within the triangle. All cesium compounds are located on the edge between the Cs and CsF, and all the fluorine compounds are located on the edge between CsF and F₂. The compounds of all other elements are inside the triangle. The position of the compound on or in the triangle defines its bonding character. The closer the position of the compound toward the Cs corner, the more metallic, the closer the position is to the F₂ corner, the more covalent, and the closer it is to the CsF corner, the more ionic. For example Li₂O is located close to CsF, and the bonding would be predominantly ionic, although there is also a small degree of covalent and metallic bonding. In contrast to that, SrMg has mostly metallic bonding, a little bit of covalent bonding, and even less ionic bonding. The bonding situation in LiH is in between metallic and ionic, with a little bit of covalent bonding character mixed in. Note that a 100% ionic bond is not possible because the electronegativity difference is finite, and thus there must always be a certain degree of electron sharing. In contrast to that, a 100% covalent bond is possible because electrons can be equally shared between two atoms when the electronegativity difference is zero.

The take home message is that in compounds and elements there is usually a mix of bonding types, and not a single bonding type, even through one bonding type may strongly dominate. The concept of electronegativity, and Ketelaar's triangle in particular, is extremely helpful to predict to which degree the three bonding types are present in a substance.

Dr. Kai Landskron ([Lehigh University](#)). If you like this textbook, please consider to make a donation to support the author's research at Lehigh University: [Click Here to Donate](#).

This page titled [1.3: Periodic Properties of Atoms](#) is shared under a [not declared](#) license and was authored, remixed, and/or curated by [Kai Landskron](#).

- [1.3: Periodic Properties of Atoms](#) by [Kai Landskron](#) is licensed [CC BY 4.0](#).